SMART CANTEEN PRE-ORDER & QUEUE MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

THARIKA R

(241401117)

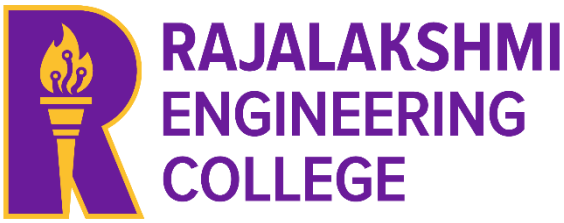
in partial fulfilment of the End Semester Examination for the course

CB23F35 – LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE AND BUSINESS SYSTEMS



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

NOVEMBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report titled **"Smart Canteen Pre-Order & Queue Management System"** is the bonafide work of **Tharika R (241401117)**, who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

SIGNATURE

Dr. K.R.Sowmia
HEAD OF THE DEPARTMENT
Department of Computer Science and Business Systems
Rajalakshmi Engineering College
Thandalam – 602 105

SIGNATURE

Mr. M.L.Alphin Ezhil Manuel
ASSISTANT PROFESSOR
Department of Computer Science and Business Systems
Rajalakshmi Engineering College

Submitted for the End Semester Examination (Individual Project) held on _____.

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, Tharika R (241401117), hereby declare that the project report titled “Smart Canteen Pre-Order & Queue Management System”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, dashboard page, Deluge script and workflows — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: _____

SIGNATURE OF THE STUDENT
THARIKA R
241401117

ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman, Chairperson and Vice-Chairman for providing the infrastructure and facilities that made this work possible.

I am grateful to our Principal, Rajalakshmi Engineering College, for the constant encouragement extended to students to take up industry-relevant learning.

I thank the Head of the Department, Department of Computer Science and Business Systems, for the guidance and support provided throughout the course.

I owe my sincere thanks to my course faculty for the valuable guidance, reviews and feedback that shaped this application and report.

I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

THARIKA R

ABSTRACT

College canteens serve a large number of students within short break periods, yet ordering, payment and collection are commonly handled manually at the counter. Students stand in a physical queue, the availability of items is not known in advance, and orders and payments are noted on paper. This causes long waiting times, errors in billing and very little visibility of canteen activity for the management.

This project presents the Smart Canteen Pre-Order & Queue Management System, developed on the Zoho Creator low-code platform. The application contains six forms – Student Registration, Food Menu, Food Order, Order Items, Payment Details and Queue Token Management – connected through lookup fields so that every order, ordered item, payment and queue token can be traced to a registered student. Reports display and manage the records of the forms, and a custom dashboard built with the Zoho Creator UI Builder presents KPI cards and charts for monitoring canteen operations. Deluge scripting and form workflows calculate the subtotal of each order item automatically as Quantity × Unit Price when either value is entered.

At the time of reporting, the data model, forms, reports, dashboard and subtotal automation have been configured, whereas automatic queue token generation is still being configured and has not yet been tested. Integration with other Zoho services, role-based access control and deployment to a production environment are identified as future work. The report documents the requirement analysis, design, implementation, planned test cases and limitations of the current version, and shows how a low-code approach allows a working canteen application to be built by a student developer without a conventional coding stack.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Canteen Pre-Order; Queue Management*

PROJECT AT A GLANCE

Attribute	Details
Project Title	Smart Canteen Pre-Order & Queue Management System
Application Domain	Education – College Canteen Management
Student Name & Register No.	Tharika R – 241401117
Zoho Creator Edition	Zoho Creator (course laboratory account)
Application Name (Link Name)	Smart Canteen Pre-Order & Queue Management System
Published / Live URL	Accessed through the developer's Zoho Creator account
Demo Credentials (per role)	Administrator, Canteen Staff and Student (RBAC configuration)
Number of Forms / Subforms	6 / 0
Number of Reports (by type)	6 List reports for the application forms
Number of Pages / Dashboards	1 custom dashboard page (UI Builder) with KPI cards and charts
Number of Workflows	8 form workflows; no scheduled workflows, approvals, blueprints or batch workflows
Deluge Functions / Custom Actions	5 Deluge scripts for subtotal calculation, order total calculation, food availability validation, queue token generation and payment status update
Integrations	Zoho Cliq integration
Roles & Profiles Defined	Administrator, Canteen Staff and Student (planned)
Environments Used	Development version of the application; Stage and Production deployment planned
Date of Production Deployment	Not yet deployed to production

TABLE OF CONTENTS

ABSTRACT iv
PROJECT AT A GLANCE v
LIST OF TABLES ix
LIST OF FIGURES x
LIST OF ABBREVIATIONS xi
INTRODUCTION 12
1.1 Background 12
1.2 Problem Statement 12
1.3 Objectives 12
1.4 Scope of the Project 13
1.5 Justification for a Low-Code Approach 13
1.6 Organisation of the Report 13
LITERATURE SURVEY AND SYSTEM STUDY 14
2.1 Study of the Existing System 14
2.2 Limitations of the Existing System 14
2.3 Low-Code versus Traditional Development 15
2.4 Literature Survey 15
2.5 Emerging Trends in Low-Code Development 16
2.6 Proposed System 16
REQUIREMENT ANALYSIS 17
3.1 Stakeholders and User Roles 17
3.2 Functional Requirements 17
3.3 Non-Functional Requirements 18
3.4 Hardware and Software Requirements 18
3.5 Use Case Model 19
SYSTEM DESIGN 20
4.1 System Architecture 20
4.2 Data Model 21
4.3 Form Design 22
4.4 Process and Workflow Design 24
4.5 User Interface and Navigation Design 24
4.6 Access Control Design 25
IMPLEMENTATION 26
5.1 Application Setup 26
5.2 Forms and Data Capture 26

5.3 Reports 30
5.4 Pages and Dashboards 33
5.5 Deluge Scripting 34
5.6 Workflows and Automation 35
5.7 Integrations 36
5.8 User Management and Role-Based Access 37
5.9 AI and Analytics Features 38
5.10 Pro-Code Extension 39
5.11 Mobile and Portal Access 39
TESTING AND DEBUGGING 40
6.1 Testing Strategy 40
6.2 Test Cases 40
6.3 Debugging Deluge Scripts 41
6.4 Defect Log 42
6.5 User Acceptance Testing 42
DEPLOYMENT, SECURITY AND GOVERNANCE 43
7.1 Environments and Deployment Strategy 43
7.2 Performance Optimisation 44
7.3 Security Features 44
7.4 Governance and Administration 44
RESULTS AND DISCUSSION 46
8.1 Application Output 46
8.2 Achievement of Objectives 47
8.3 Comparison with the Existing System 47
8.4 Limitations of the Low-Code Approach 48
8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum 48
CONCLUSION AND FUTURE ENHANCEMENTS 49
9.1 Conclusion 49
9.2 Future Enhancements 49
9.3 Learning Outcomes and Career Relevance 49
REFERENCES 50
APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST 51
APPENDIX B – COURSE OUTCOME MAPPING 53
APPENDIX C – COMPLETE DELUGE SOURCE CODE 54
APPENDIX D – USER MANUAL 2
D.1 Accessing the Application 2
D.2 Instructions for Students 2

D.3 Instructions for Canteen Staff and Administrator 2
APPENDIX E – EVALUATION RUBRIC 3
APPENDIX F – VIVA-VOCE RECORD SHEET 5

LIST OF TABLES

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application 15
Table 2.2: Summary of Literature Survey 15
Table 3.1: Stakeholders and User Roles 17
Table 3.2: Functional Requirements 17
Table 3.3: Non-Functional Requirements 18
Table 3.4: Hardware and Software Requirements 18
Table 3.5: Use Case Description – Place Food Pre-Order and Add Order Items 19
Table 4.1: Entities and Relationships 21
Table 4.2: Field Specification – Student Registration Form 22
Table 4.3: Field Specification – Food Menu Form 22
Table 4.4: Field Specification – Food Order Form 22
Table 4.5: Field Specification – Order Items Form 23
Table 4.6: Field Specification – Payment Details Form 23
Table 4.7: Field Specification – Queue Token Management Form 23
Table 4.8: Role–Permission Matrix (Proposed Design) 25
Table 5.1: Reports Implemented 30
Table 5.2: Deluge Scripts Implemented 34
Table 5.3: Workflows Implemented 36
Table 5.4: Integrations Implemented 36
Table 6.1: Test Cases 40
Table 6.2: Defect Log 42
Table 6.3: User Acceptance Feedback 42
Table 7.1: Deployment Checklist 43
Table 7.2: Performance Optimisation Results 44
Table 7.3: Security Controls Implemented 44
Table 8.1: Mapping of Objectives to Outcomes 47
Table 8.2: Existing System versus Proposed Application 47
Table A.1: Feature Coverage Checklist 51
Table B.1: Mapping of Course Outcomes to Report Chapters 53
Table E.1: Evaluation Rubric for the Individual Project 3
Table E.2: Performance Level Descriptors 4
Table F.1: Viva-Voce Questions and Responses 5
Table F.2: Marks Summary 5

LIST OF FIGURES

Figure 2.1: As-Is Process Flow of the Existing System 14
Figure 3.1: Use Case Diagram of the Application 19
Figure 4.1: System Architecture of the Application 20
Figure 4.2: Entity–Relationship Diagram 21
Figure 4.3: Workflow / Process Flow for Food Pre-Order and Order Items 24
Figure 4.4: Navigation Structure and Wireframe of the Dashboard Page 24
Figure 5.1: Application Home and Navigation Sections 26
Figure 5.2: Order Items Form in the Form Builder 27
Figure 5.3: Order Items Form in Live Mode 28
Figure 5.4: Food Order Report 33
Figure 5.5: Smart Canteen Dashboard Page (UI Builder) 34
Figure 5.6: Order Items Subtotal Workflow Configuration 36
Figure 8.1: Scenario Step 1 – Student Registration and Food Order Creation 46
Figure 8.2: Scenario Step 2 – Order Item Entry with Automatic Subtotal 46
Figure 8.3: Scenario Step 3 – Dashboard Monitoring of Canteen Operations 47

LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
Deluge	Data Enriched Language for the Universal Grid Environment
ER	Entity–Relationship
FR	Functional Requirement
KPI	Key Performance Indicator
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
REST	Representational State Transfer
UAT	User Acceptance Testing
UI	User Interface

CHAPTER 1
INTRODUCTION

1.1 Background

A college canteen is one of the busiest service points on a campus. Students have only short break intervals, and many of them reach the canteen at the same time. In a typical canteen the student joins a queue, selects items from a displayed menu, places the order verbally, pays, and waits for the order to be called out. Orders and amounts are noted on paper, and the management has no consolidated view of what has been ordered, paid for and collected.

The Smart Canteen Pre-Order & Queue Management System addresses this process. It is built as the individual project for the course CB23F35 – Low-Code Application Development using Zoho Creator, and covers student registration, the food menu, pre-orders with order items, payments and queue tokens, with reports and a dashboard for the canteen staff.

1.2 Problem Statement

Manual ordering at the canteen counter leads to long queues, loss of break time and errors in order and payment records, and gives the management no timely view of orders, payments and waiting customers. A digital pre-order and token-based queue system is therefore required.

- Long waiting time at the counter in peak hours, as every order is taken in person.
- No advance ordering and no visibility of item availability.
- Manual order, amount and payment records that are error-prone and hard to verify.
- No real-time visibility of orders, payments and the queue for the management.

1.3 Objectives

- To design a normalised data model with six interlinked forms for students, menu, orders, order items, payments and queue tokens.
- To automate the item subtotal (Quantity × Unit Price) using Deluge scripting and workflows.
- To provide reports and a UI Builder dashboard with KPI cards and charts for monitoring operations.
- To configure automatic queue token generation (work in progress).
- To identify integration, role-based access control and production deployment as future enhancements.

1.4 Scope of the Project

In scope: six forms – Student Registration, Food Menu, Food Order, Order Items, Payment Details and Queue Token Management – related through lookup fields; reports for their records; a UI Builder dashboard; and Deluge scripting with workflows for the automatic subtotal. The application is used through the Zoho Creator web interface.

Out of scope for this version: automatic queue token generation (still being configured), integration with other Zoho services or payment systems, role-based access control, a customer portal and production deployment.

1.5 Justification for a Low-Code Approach

A low-code application platform (LCAP) lets an application be assembled through visual configuration with limited hand-written code, which shortens delivery and widens the group of people who can build business applications [1]. Practitioners report that its drag-and-drop interfaces and ready-made components make it easy to learn and speed up development, and that it is favoured where workflows need automation [2]. This suits a canteen application made of forms, relationships between records, reports and a few calculations. Zoho Creator provides forms, lookups, reports, a page builder, Deluge, workflows and hosting in one platform, so the project could focus on the data model and business rules.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing manual system the student joins the queue, checks the menu board, places the order verbally at the counter, and the staff note the order and collect the payment. The food is prepared while the student waits and the order is finally called out (Figure 2.1). Every step requires physical presence in the queue, and no digital record of the order or payment is created.

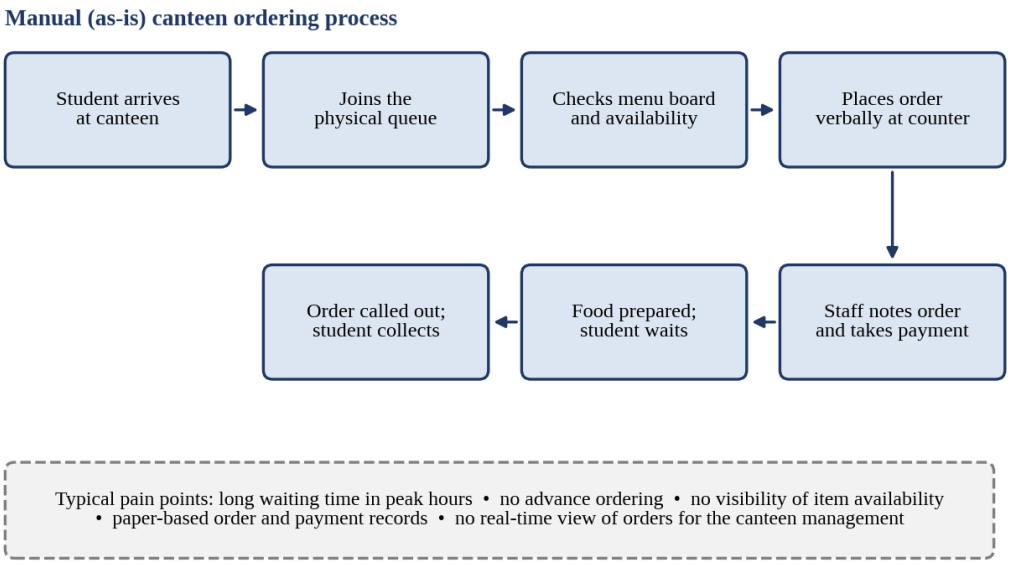


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Long queues and waiting time at the counter in peak periods, because ordering, billing and serving happen at one point.
- No advance (pre-)ordering and no visibility of item availability before reaching the counter.
- Manual and paper-based records of orders, item amounts and payments are error-prone and cannot be searched or summarised easily.
- No consolidated reports or dashboard, and no means to track an order or a queue position after it is placed.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Separate design and coding of front end, back end and database	Forms, reports and dashboard built visually; Deluge only for the subtotal
Skills required	Front-end, server and database programming	Data modelling, form design, basic Deluge
Hosting & infrastructure	Server and database set-up needed	Provided by the Zoho Creator cloud
Mobile application	Separate mobile client needed	Mobile access through Zoho Creator; not evaluated here
Security & access control	Authentication and authorisation coded by hand	Platform sign-in; role permissions not yet configured
Maintenance & change	Code change, rebuild, redeploy	Fields, forms and reports edited in the builder
Customisation limits	Practically unrestricted	Limited to platform components and Deluge
Cost	Developer effort plus hosting	Platform subscription; no server cost

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Bock & Frank, 2021 [1]	Low-Code Platform	Describes low-code platforms as development environments linked to citizen development, agility and productivity.	Conceptual basis for choosing low-code.
2	Luo et al., 2021 [2]	Characteristics and Challenges of Low-Code Development	From 301 developer-forum posts: drag-and-drop development with little code, faster delivery, suited to workflows; views on limits differ.	Supports a workflow-centred app; informs Section 8.4.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
3	Waszkowski, 2019 [3]	Low-code platform for automating business processes in manufacturing	Uses the Aurea BPM low-code platform to automate business processes with minimal coding.	Comparable to automating the order process.
4	Sahay et al., 2020 [4]	Supporting the understanding and comparison of low-code development platforms	Compares low-code platforms on a set of features.	Guides platform feature comparison.
5	Zoho Corp. [5]	Creator Help: Form workflow actions	Form workflows run actions, including Deluge scripts, on events such as user input of a field.	Basis of the subtotal workflow.

2.5 Emerging Trends in Low-Code Development

Current low-code practice is shaped by citizen development, hyper-automation of repeated manual steps, and AI-assisted building features [1], [2]. The canteen application follows the first two trends: it is built visually by a student developer and automates the manual item calculation and, in future, queue token issue. AI-assisted features are not part of the implemented application (Section 5.9).

2.6 Proposed System

The proposed system is a Zoho Creator application in which students are registered once and food orders are captured digitally as pre-orders linked to a student through a lookup. Ordered items carry quantity, unit price and an automatically calculated subtotal, which removes manual calculation. Payments and queue tokens are linked to the order, and the Food Menu keeps prices and availability in one place. Reports and a dashboard give the management an up-to-date view, overcoming the lack of records in Section 2.2; automatic queue token generation, which addresses waiting time, is still being configured.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Canteen manager or application owner	Configure the application; view reports and dashboard
Canteen Staff	Staff who prepare food and run the counter	Maintain menu; verify orders; record payments and tokens
Student	College student who orders food	Register; view menu; place orders; collect with token

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall register students with name, register number, department and contact details.	Student, Staff	High
FR-02	The system shall maintain food items with category, price and availability.	Staff	High
FR-03	The system shall create a food pre-order for a registered student through a lookup.	Student	High
FR-04	The system shall add ordered items to an order by selecting menu items with quantity and unit price.	Student	High
FR-05	The system shall compute Subtotal = Quantity × Unit Price when either value is entered in Order Items.	System	High
FR-06	The system shall record the payment details of a food order.	Staff	High
FR-07	The system shall maintain a queue token for each order; automatic generation is under configuration.	System, Staff	High
FR-08	The system shall provide reports to display and manage the records of every form.	Staff, Admin	Medium
FR-09	The system shall provide a dashboard with KPI cards and charts for monitoring operations.	Admin	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Forms and reports should open within a few seconds on a normal campus internet connection.
NFR-02	Security	Only authenticated Zoho users shall access the application; role-based restriction of data is planned.
NFR-03	Usability	The application shall be usable in a standard web browser, with lookups and drop-downs limiting typing.
NFR-04	Availability	The application is hosted on the Zoho Creator cloud, so its availability depends on the platform.
NFR-05	Maintainability	Fields, reports and workflows shall be changeable in the builder without redeploying code.
NFR-06	Auditability	Changes to records shall be traceable through the audit facilities of Zoho Creator.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (course laboratory account)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
API testing tool	Not used in this version (no REST API integration implemented)
Other Zoho services	None used in the current version
Mobile	Not evaluated in this version
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

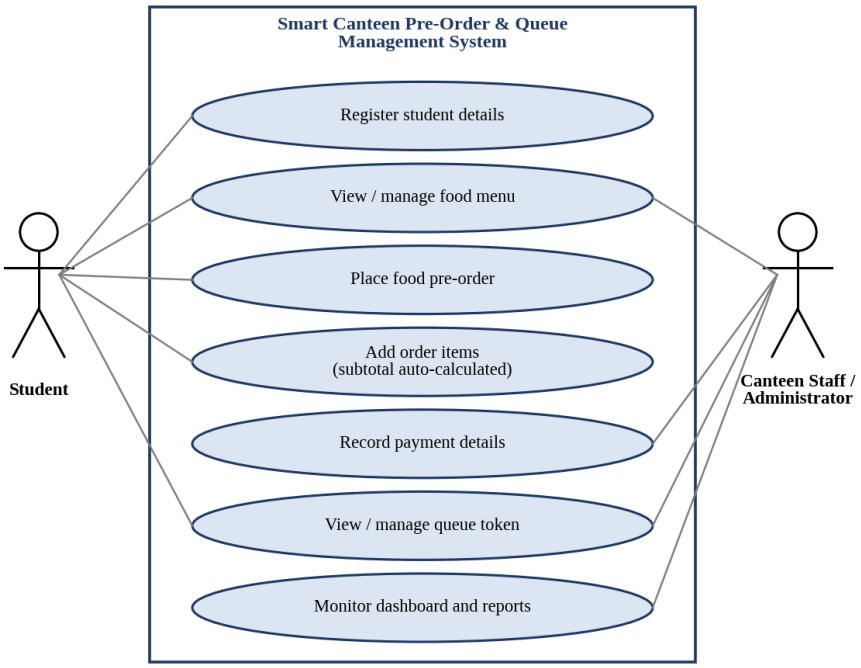


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Place Food Pre-Order and Add Order Items

Use Case ID	UC-01
Use Case Name	Place Food Pre-Order and Add Order Items
Actor(s)	Student
Pre-condition	The student is registered in Student Registration, and the required items exist in Food Menu.
Main Flow	1. The student opens the Food Order form and selects the registered student through the lookup. 2. The food order is submitted. 3. In the Order Items form, the student selects the Food Order and the food item through lookups. 4. The student enters Quantity and Unit Price. 5. The workflow calculates Subtotal = Quantity × Unit Price. 6. The order item is submitted.
Alternate Flow	If the student is not yet registered, the student is first added through the Student Registration form.
Post-condition	The Food Order and Order Items records are stored, each item with its calculated subtotal.
Related FR	FR-03, FR-04, FR-05

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

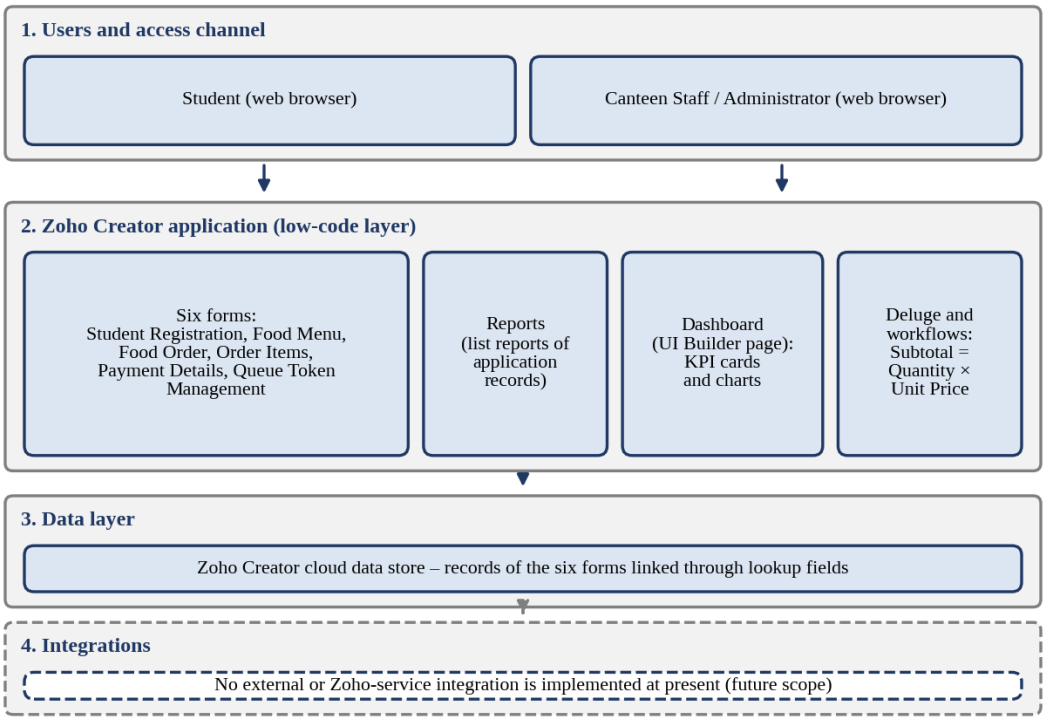


Figure 4.1: System Architecture of the Application

The application has four layers (Figure 4.1). Students and canteen staff use a web browser. The Zoho Creator layer holds the six forms, their reports, the UI Builder dashboard, and the Deluge script and workflows for the item subtotal. Records are stored in the Zoho Creator cloud and linked through lookup fields. No integration with another Zoho service or external API is implemented, so that layer is shown as future scope.

4.2 Data Model

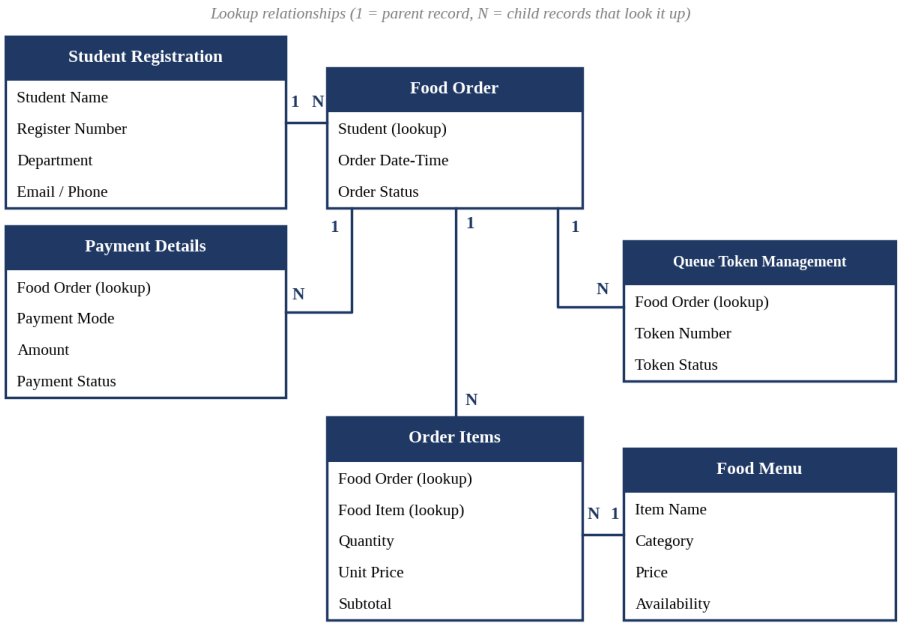


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Student Registration	Student Name, Register Number, Department, Email, Phone Number	Food Order	One-to-Many	Lookup in Food Order
Food Menu	Item Name, Category, Price, Availability	Order Items	One-to-Many	Lookup in Order Items
Food Order	Student, Order Date-Time, Order Status	Order Items, Payment Details, Queue Token Management	One-to-Many	Lookup in each child form
Order Items	Food Order, Food Item, Quantity, Unit Price, Subtotal	Food Order, Food Menu	Many-to-One	Lookup fields
Payment Details	Food Order, Payment Mode, Amount, Payment Status	Food Order	Many-to-One	Lookup field
Queue Token Management	Food Order, Token Number, Token Status	Food Order	Many-to-One	Lookup field

Relationships are realised only through lookup fields; no subform is used. A student can place many orders, an order can contain many order items, and each order item refers to one menu item. One payment record and one token are expected per order in normal operation.

4.3 Form Design

Each form is specified below; link names replace the spaces of the field label with underscores.

Table 4.2: Field Specification – Student Registration Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Student Name	Student_Name	Single Line	Yes	No specific validation configured	Full name of the student
Register Number	Register_Number	Single Line	Yes	No specific validation configured	Identifies the student
Department	Department	Single Line	No	No specific validation configured	Department of study
Email	Email	Email	No	No specific validation configured	Contact detail
Phone Number	Phone_Number	Phone	No	No specific validation configured	Contact detail

Table 4.3: Field Specification – Food Menu Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Item Name	Item_Name	Single Line	Yes	Cannot be empty	Name of the food item
Category	Category	Drop Down	No	Predefined options	Type of item
Price	Price	Decimal	Yes	Greater than 0	Unit price of the item
Availability	Availability	Drop Down	No	Available / Not Available	Available / not available

Table 4.4: Field Specification – Food Order Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Student	Student	Lookup	Yes	Select registered student	Lookup to Student Registration

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Order Date-Time	Order_Date_Time	Date-Time	No	Current date-time (if configured)	Time of the pre-order
Order Status	Order_Status	Drop Down	No	Pending (if configured)	Status of the order

Table 4.5: Field Specification – Order Items Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Food Order	Food_Order	Lookup	Yes	Select existing record	Lookup to Food Order
Food Item	Food_Item	Lookup	Yes	Select existing record	Lookup to Food Menu
Quantity	Quantity	Number	Yes	Greater than 0	Triggers subtotal workflow
Unit Price	Unit_Price	Decimal	Yes	Greater than 0	Triggers subtotal workflow
Subtotal	Subtotal	Decimal	Yes	Quantity × Unit Price	Set by Deluge script

Table 4.6: Field Specification – Payment Details Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Food Order	Food_Order	Lookup	Yes	Select existing record	Lookup to Food Order
Payment Mode	Payment_Mode	Drop Down	No	Predefined options	Mode of payment
Amount	Amount	Decimal	No	Greater than 0	Amount paid
Payment Status	Payment_Status	Drop Down	No	Predefined options	Payment tracking

Table 4.7: Field Specification – Queue Token Management Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Food Order	Food_Order	Lookup	Yes	Select existing record	Lookup to Food Order
Token Number	Token_Number	Single Line	No	Auto-generated	Automatic generation under configuration
Token Status	Token_Status	Drop Down	No	Waiting (if configured)	Status of the token

4.4 Process and Workflow Design

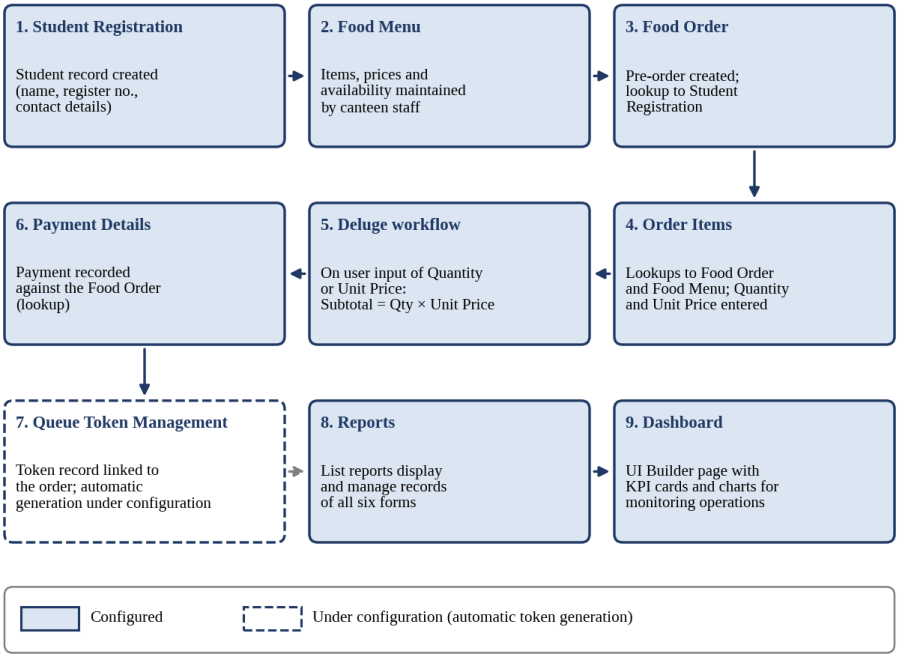


Figure 4.3: Workflow / Process Flow for Food Pre-Order and Order Items

The Deluge script and workflow run in the Order Items form when Quantity or Unit Price is entered. Automatic generation of the queue token is outlined with a dashed line because it is still being configured and has not been tested.

4.5 User Interface and Navigation Design

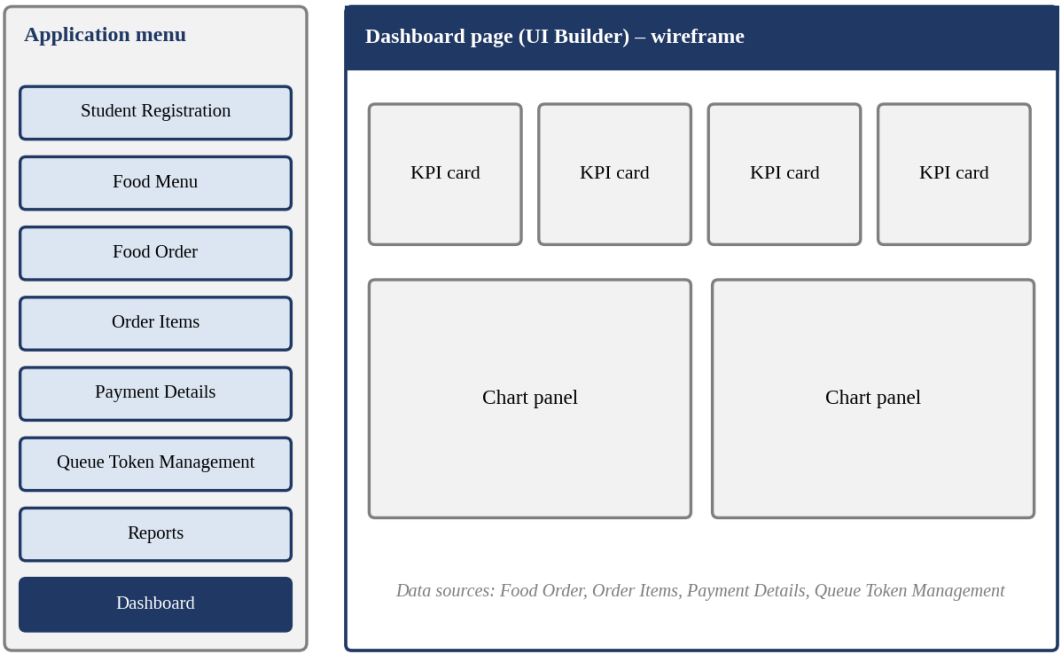


Figure 4.4: Navigation Structure and Wireframe of the Dashboard Page

Figure 4.4 shows the intended navigation, with one menu entry per form, the reports and the dashboard, and the dashboard wireframe. It is a design wireframe; the implemented page is shown in Figure 5.5.

4.6 Access Control Design

Table 4.8: Role–Permission Matrix (Proposed Design)

Component	Administrator	Canteen Staff	Student
Form: Student Registration	A E V D	A E V	A V (own)
Form: Food Menu	A E V D	A E V	V
Form: Food Order	A E V D	E V	A V (own)
Form: Order Items	A E V D	E V	A V (own)
Form: Payment Details	A E V D	A E V	V (own)
Form: Queue Token Management	A E V D	A E V	V (own)
Report: All records	V	V	—
Page: Dashboard	V	V	—

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application “Smart Canteen Pre-Order & Queue Management System” was created in Zoho Creator to hold the six forms, their reports and the dashboard page. The application menu gives access to the forms and their reports, and the dashboard page provides an overview of canteen operations. The sections that follow describe each component as it is configured in the application; features that are not yet configured are stated as such.

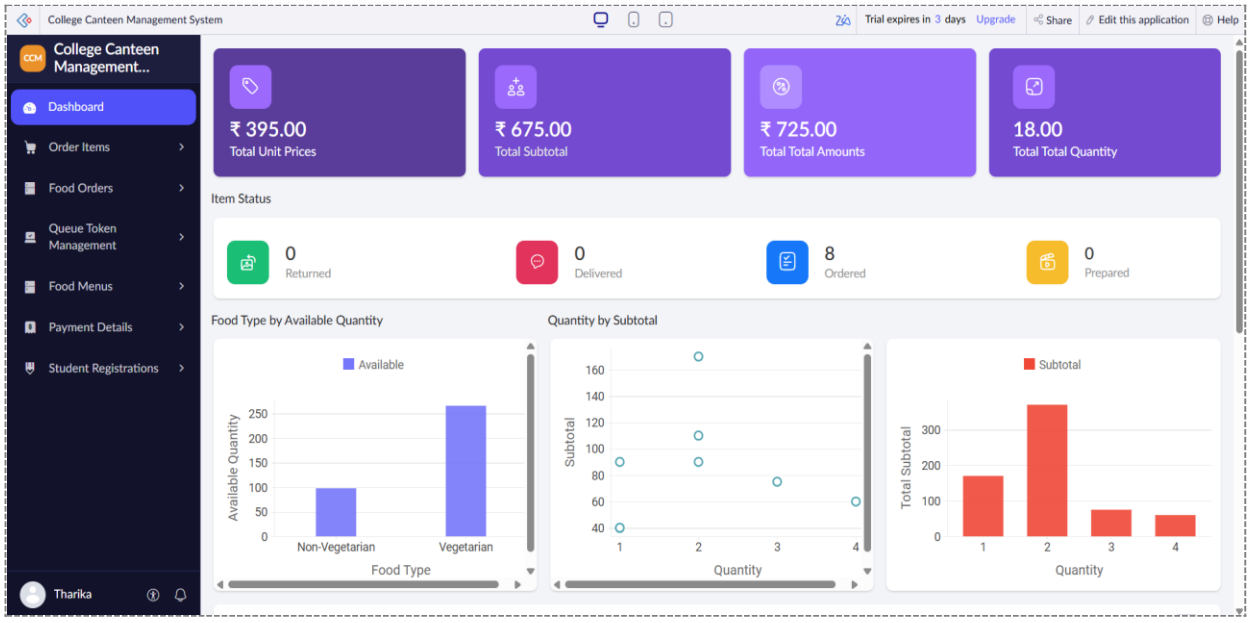


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

Six forms implement the main canteen operations. They are related through lookup fields; no subform is used.

- **Student Registration:** captures the student’s name, register number, department and contact details. It is the master record to which every food order is linked.
- **Food Menu:** stores each food item with its category, price and availability, so that prices are maintained in one place.
- **Food Order:** records a pre-order. Its Student field is a lookup to Student Registration, which ties the order to a registered student.

- **Order Items:** stores the items of an order. Food Order and Food Item are lookups to Food Order and Food Menu; Quantity and Unit Price are entered, and Subtotal is calculated by a Deluge script and workflows (Section 5.5).
- **Payment Details:** tracks the payment of an order through a lookup to Food Order, with the payment mode, amount and status.
- **Queue Token Management:** keeps the token of an order through a lookup to Food Order. Automatic token generation is still being configured and is not claimed as complete.

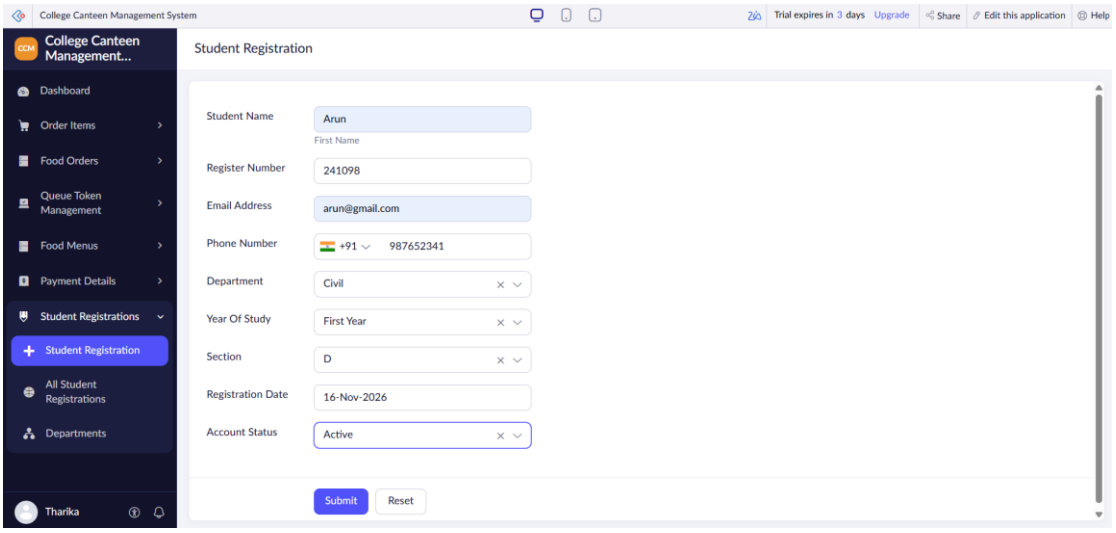


Figure 5.2(a): Student Registration Form in Live Mode

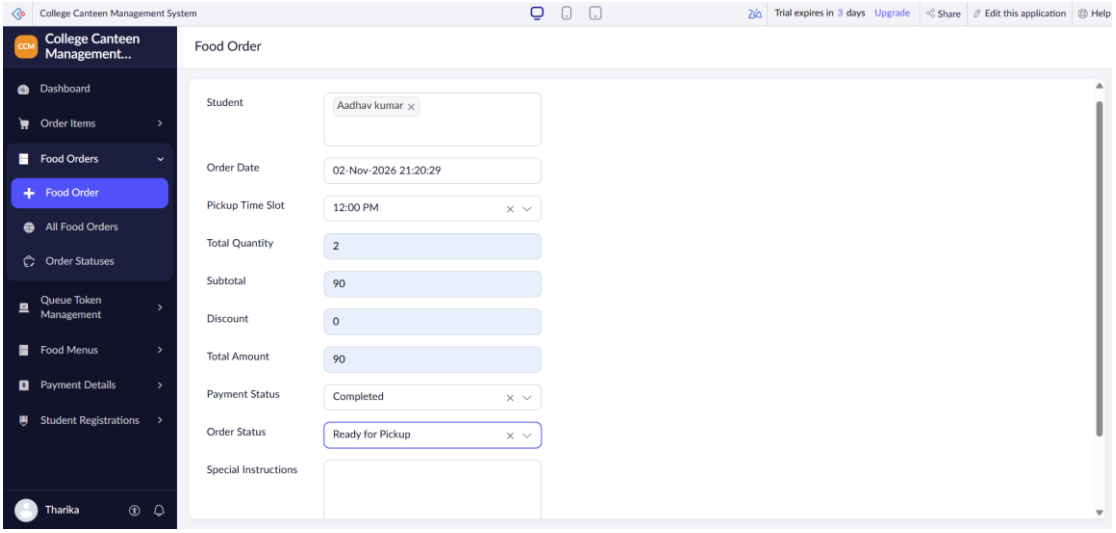


Figure 5.2(b): Food Order Form in the Live Mode

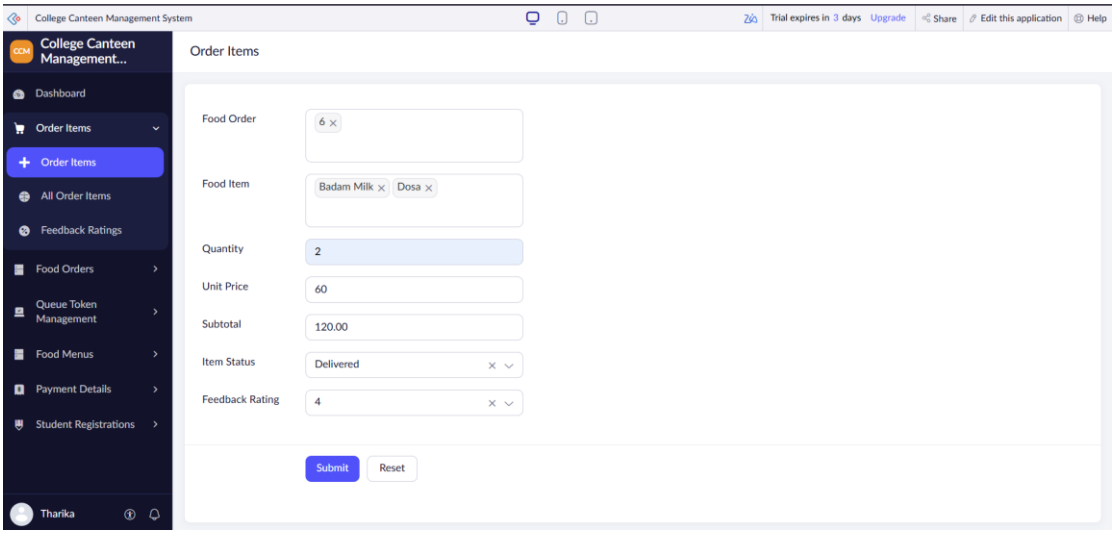


Figure 5.2(c): Order Items Form in Live Mode

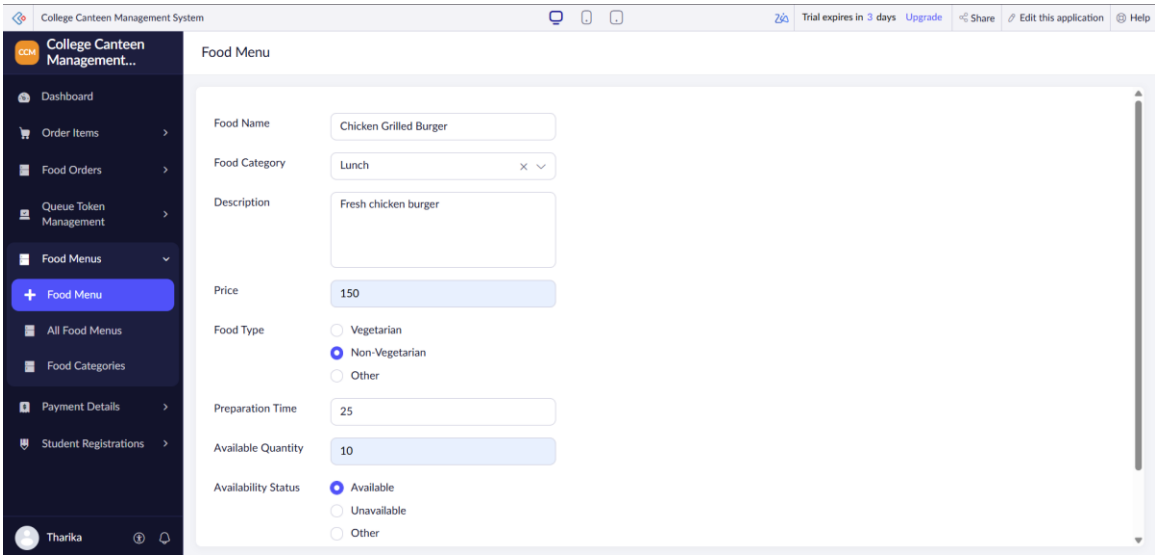


Figure 5.2(d): Food Menu Form in Live Mode

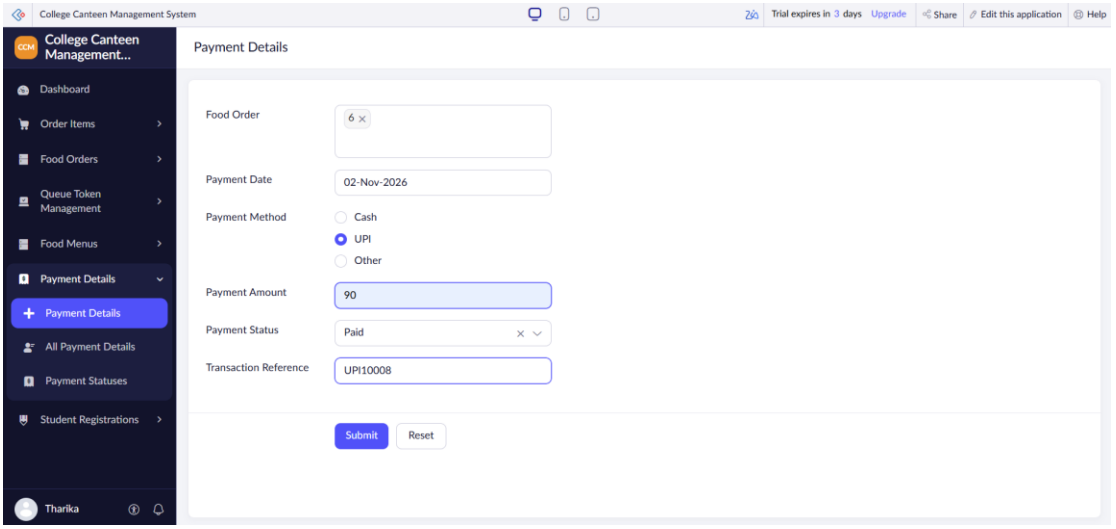


Figure 5.2(e): Payment Details Form in Live Mode

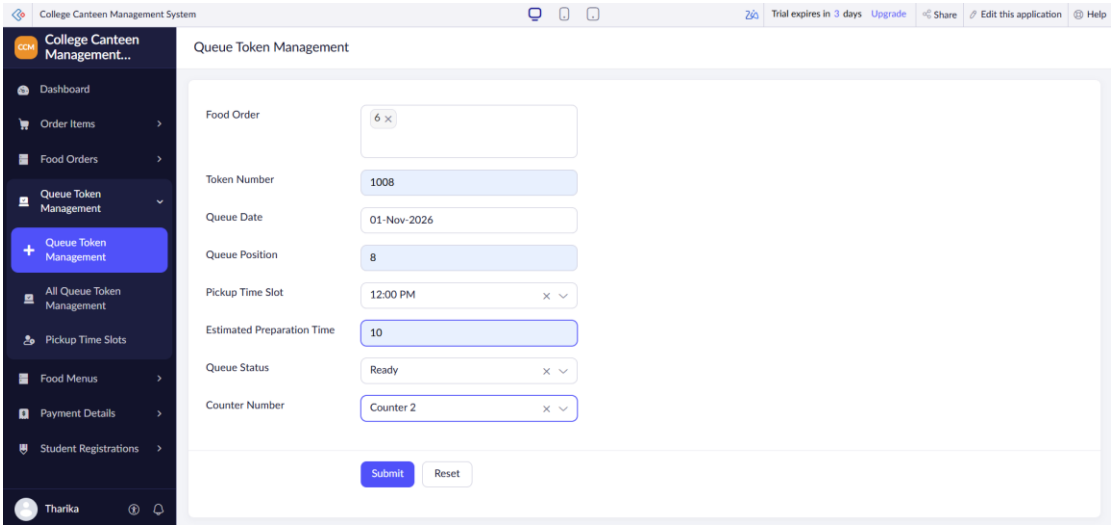


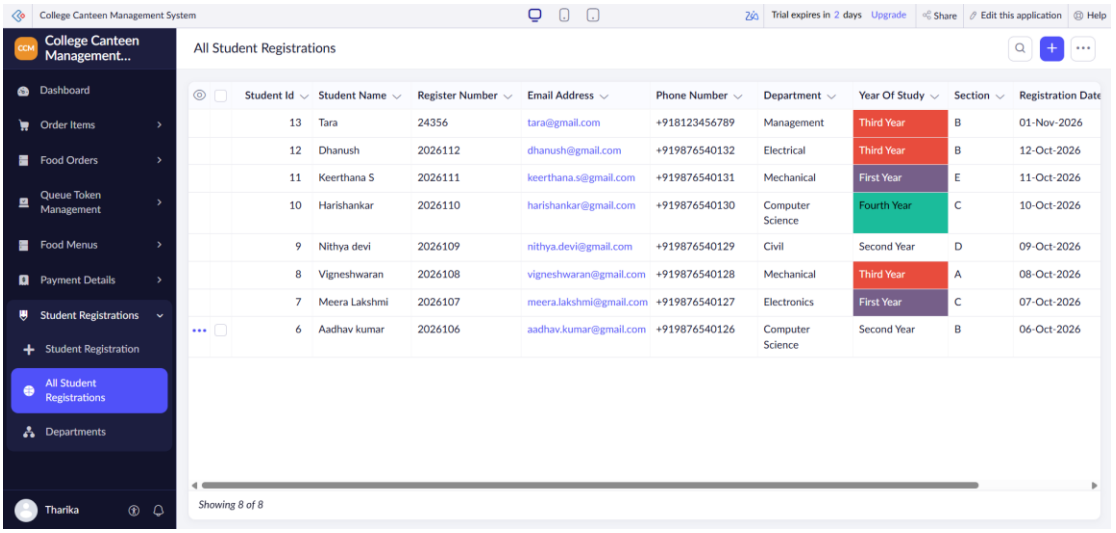
Figure 5.2(f): Queue Token Management Form in Live Mode

5.3 Reports

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Student Registration Report	List	Student Registration	Default list layout	None	All application users
Food Menu Report	List	Food Menu	Default list layout	None	All application users
Food Order Report	List	Food Order	Default list layout	None	All application users
Order Items Report	List	Order Items	Default list layout	None	All application users
Payment Details Report	List	Payment Details	Default list layout	None	All application users
Queue Token Management Report	List	Queue Token Management	Default list layout	None	All application users

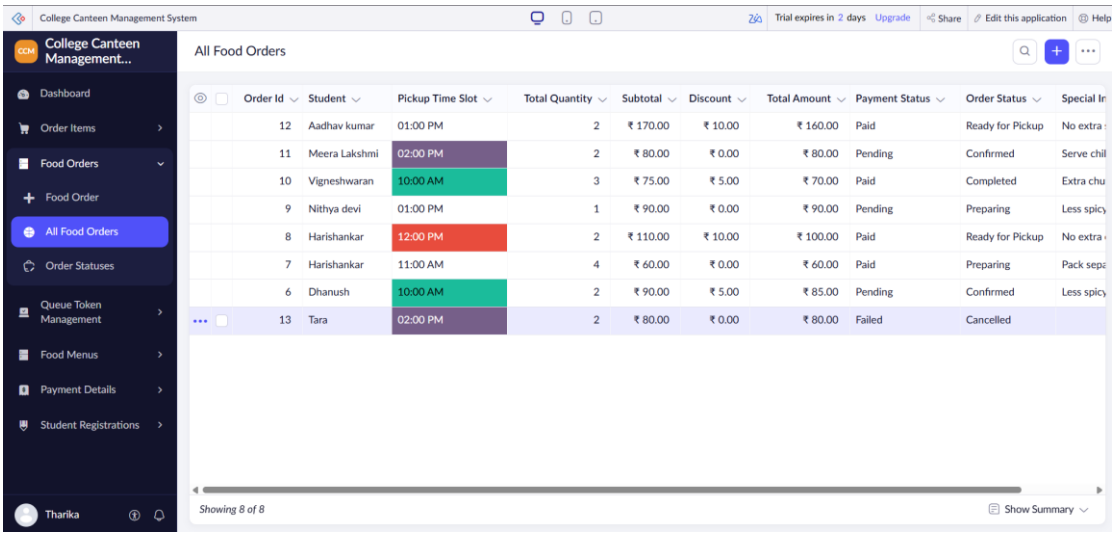
Table 5.1: Reports Implemented

Reports have been created in the Smart Canteen Pre-Order & Queue Management System to display, organize and manage the records stored in the application. Each report is associated with a specific form, allowing staff to access important information without opening individual forms separately. The reports provide a convenient way to view student registration details, available food items, food orders, order items, payment information and queue token records. They help staff monitor incoming orders, check order statuses, verify payment details, identify ordered food items and manage the queue efficiently. By presenting the stored information in a structured tabular format, the reports make record viewing, searching and editing easier. These reports also support the overall order management process by providing quick access to the information required for daily canteen operations.



Student Id	Student Name	Register Number	Email Address	Phone Number	Department	Year Of Study	Section	Registration Date
13	Tara	24356	tara@gmail.com	+918123456789	Management	Third Year	B	01-Nov-2026
12	Dhanush	2026112	dhanush@gmail.com	+919876540132	Electrical	Third Year	B	12-Oct-2026
11	Keerthana S	2026111	keerthana.s@gmail.com	+919876540131	Mechanical	First Year	E	11-Oct-2026
10	Harishankar	2026110	harishankar@gmail.com	+919876540130	Computer Science	Fourth Year	C	10-Oct-2026
9	Nithya devi	2026109	nithya.devi@gmail.com	+919876540129	Civil	Second Year	D	09-Oct-2026
8	Vigneshwaran	2026108	vigneshwaran@gmail.com	+919876540128	Mechanical	Third Year	A	08-Oct-2026
7	Meera Lakshmi	2026107	meera.lakshmi@gmail.com	+919876540127	Electronics	First Year	C	07-Oct-2026
6	Aadhav kumar	2026106	aadhav.kumar@gmail.com	+919876540126	Computer Science	Second Year	B	06-Oct-2026

Figure 5.3(a): Student Registrations Report



Order Id	Student	Pickup Time Slot	Total Quantity	Subtotal	Discount	Total Amount	Payment Status	Order Status	Special In
12	Aadhav kumar	01:00 PM	2	₹ 170.00	₹ 10.00	₹ 160.00	Paid	Ready for Pickup	No extra
11	Meera Lakshmi	02:00 PM	2	₹ 80.00	₹ 0.00	₹ 80.00	Pending	Confirmed	Serve chil
10	Vigneshwaran	10:00 AM	3	₹ 75.00	₹ 5.00	₹ 70.00	Paid	Completed	Extra chu
9	Nithya devi	01:00 PM	1	₹ 90.00	₹ 0.00	₹ 90.00	Pending	Preparing	Less spicy
8	Harishankar	12:00 PM	2	₹ 110.00	₹ 10.00	₹ 100.00	Paid	Ready for Pickup	No extra
7	Harishankar	11:00 AM	4	₹ 60.00	₹ 0.00	₹ 60.00	Paid	Preparing	Pack sepa
6	Dhanush	10:00 AM	2	₹ 90.00	₹ 5.00	₹ 85.00	Pending	Confirmed	Less spicy
13	Tara	02:00 PM	2	₹ 80.00	₹ 0.00	₹ 80.00	Failed	Cancelled	

Figure 5.3(b): Food Order Report

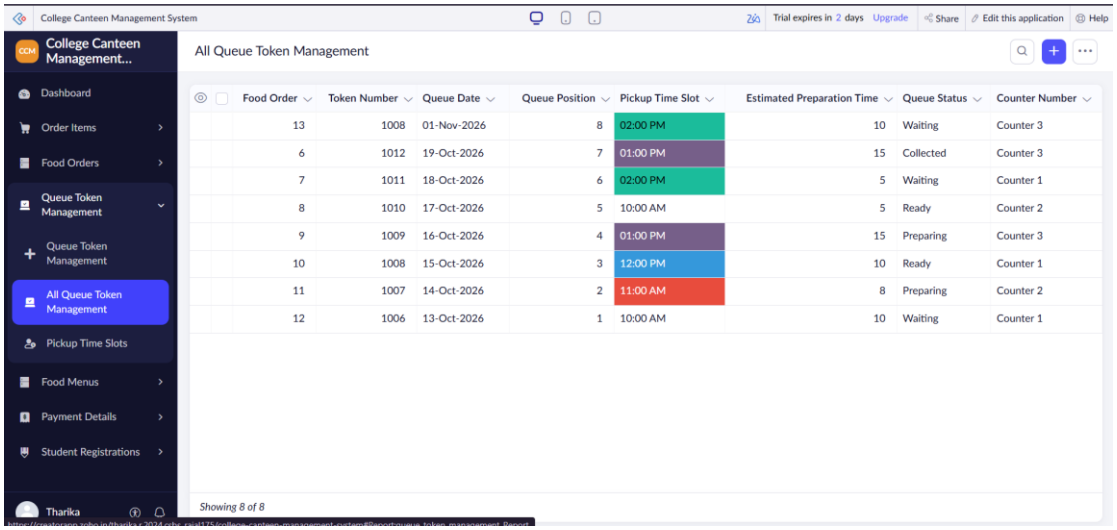


Figure 5.3(c): Queue Token Management Report

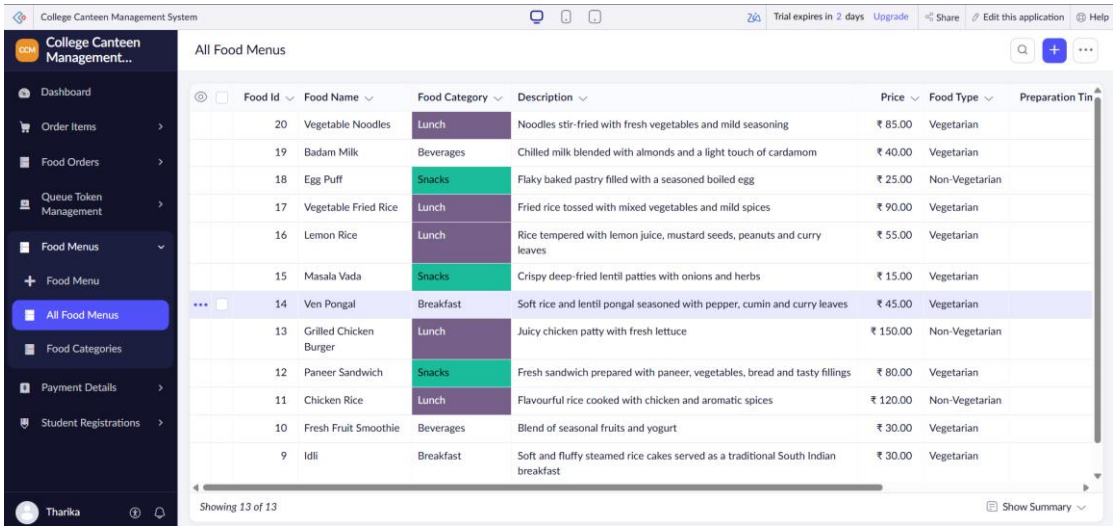


Figure 5.3(d): Food Menus Report

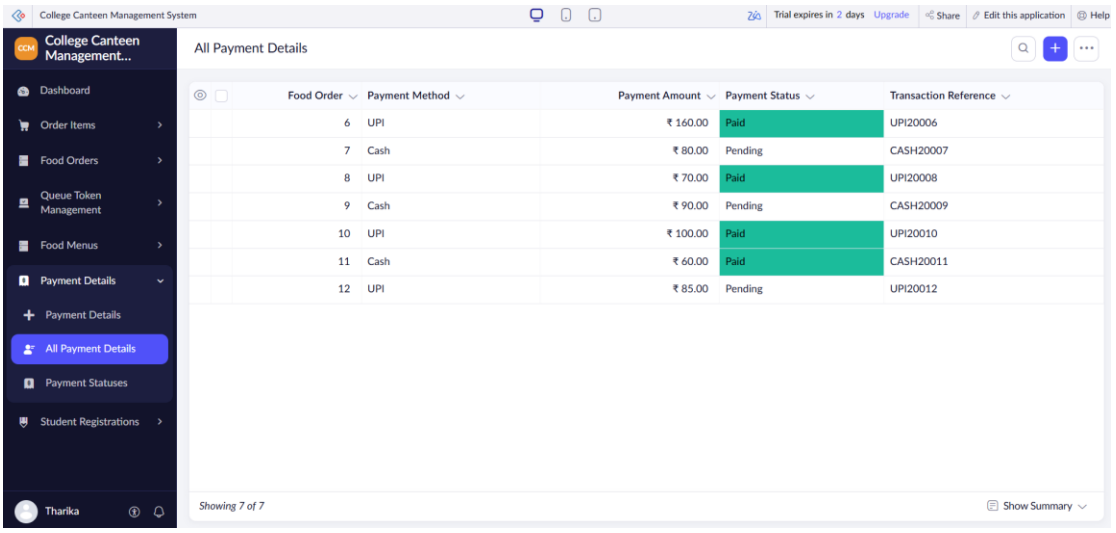


Figure 5.3(e): Payment Details Report

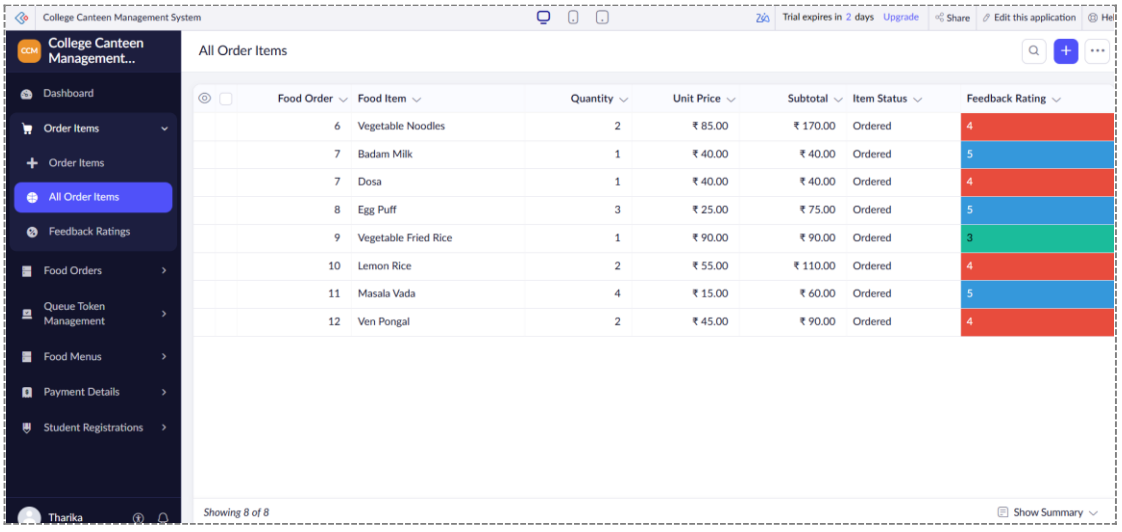


Figure 5.3(f): Order Items Report

5.4 Pages and Dashboards

A custom dashboard page was designed with the Zoho Creator UI Builder for the canteen staff and the administrator. The page combines KPI cards, which summarise canteen activity at a glance, with charts that present the application's records visually. The components draw on the data stored in the application's forms, so the dashboard reflects the orders, payments and tokens recorded in the system and helps the management monitor operations without compiling manual tallies.

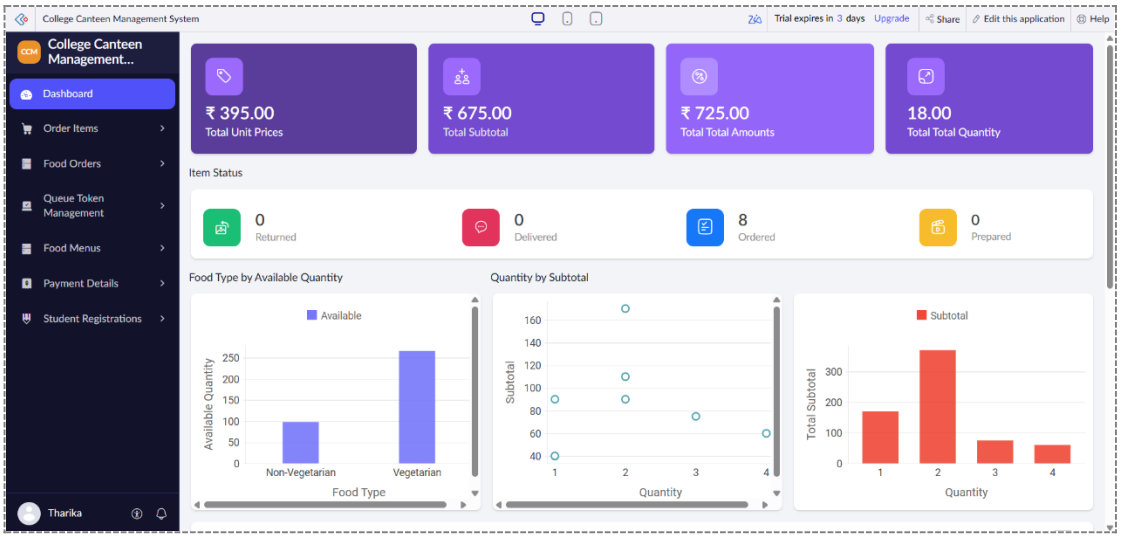


Figure 5.5(a): Smart Canteen Dashboard Page (UI Builder)

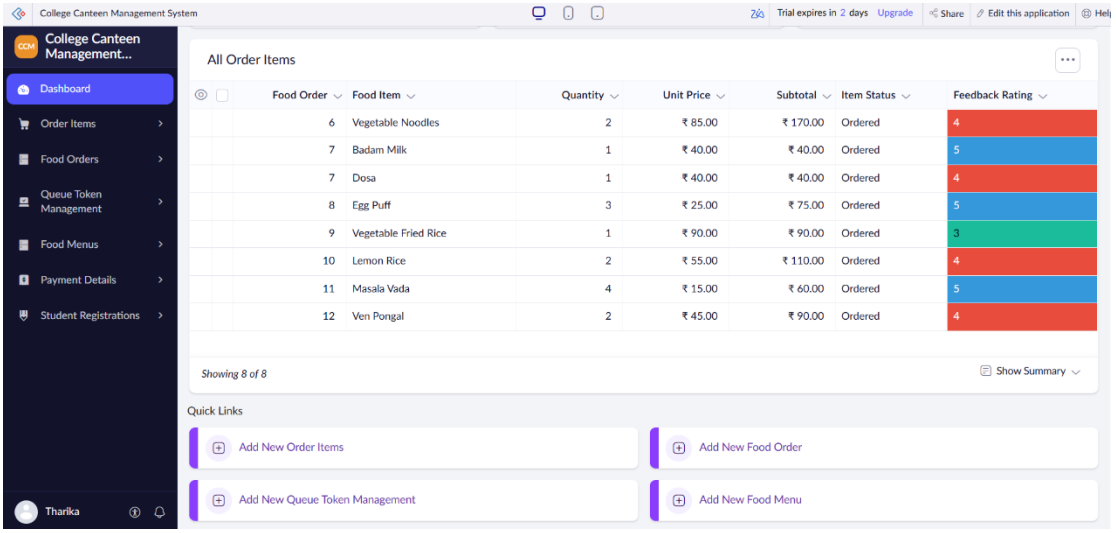


Figure 5.5(a): Smart Canteen Dashboard Page (UI Builder)

5.5 Deluge Scripting

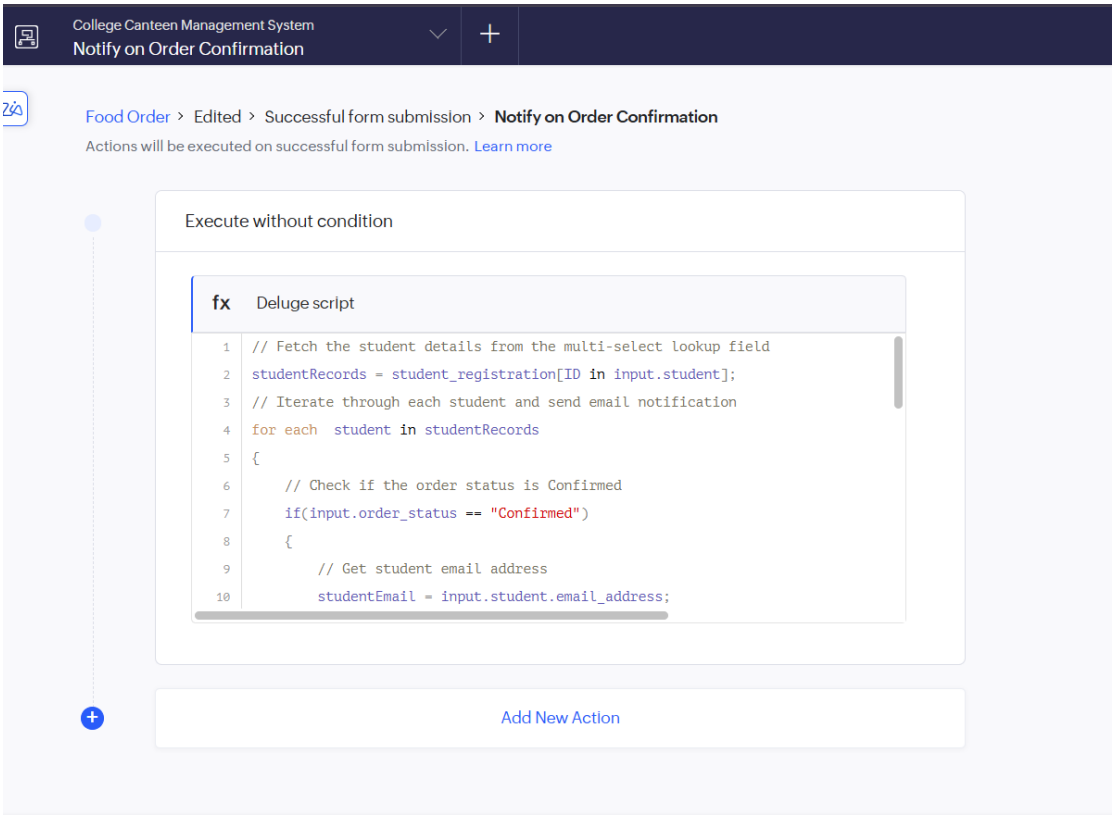
Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Order Items → field action On User Input (Quantity, Unit Price)	Calculate Subtotal = Quantity × Unit Price	input keyword, arithmetic operator, null check
DS-02	Total Order Amount Calculation	Food Order	On Success
DS-03	Food Availability and Stock Validation	Food Order	On Validate

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-04	Queue Token Generation	Queue Token Management	On Success
DS-05	Payment Status Update	Payment Details	On Success

Code 5.1: Subtotal calculation script (DS-01)

```
// DS-01 | Order Items | On User Input (Quantity, Unit Price)
// Purpose: Calculate Subtotal = Quantity × Unit Price
if (input.Quantity != null && input.Unit_Price != null)
{
    input.Subtotal = input.Quantity * input.Unit_Price;
}
```



The script runs through the On User Input event of the Quantity and Unit Price fields, which Zoho Creator executes when the value of a field changes and before the data is saved. Field values are read and set through the input keyword followed by the field link name. When both values are available, the Subtotal field is filled with their product, so the user does not calculate the amount manually

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Subtotal on Quantity	Form Workflow	On user input of Quantity (Order Items)	Quantity and Unit Price available	Set Subtotal = Quantity × Unit Price	DS-01
Subtotal on Unit Price	Form Workflow	On user input of Unit Price (Order Items)	Quantity and Unit Price available	Set Subtotal = Quantity × Unit Price	DS-01
Queue Token Generation (under configuration)	Form Workflow	To be finalised	Requested quantity exceeds available stock or food is unavailable	Generate the token for an order; not yet completed or tested	DS-03

Workflows are event-based form workflows, which can run Deluge scripts when a form event such as the user input of a field occurs [5]. Schedule, approval, Blueprint and batch workflows are not used in this version.

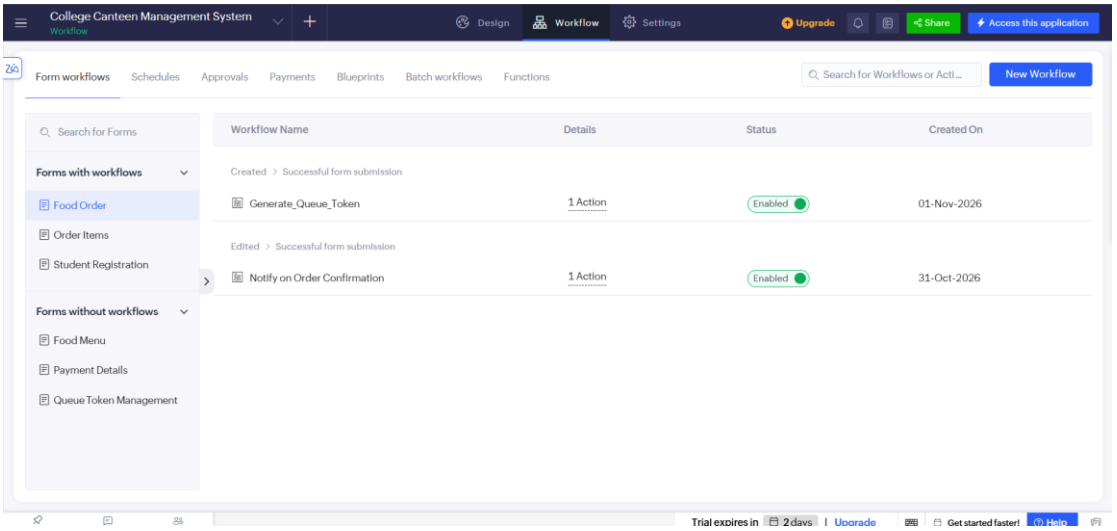


Figure 5.6: Generate Queue Token Workflow Configuration

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Zoho Cliq Notification	Zoho Cliq Messages API	POST	OAuth 2.0 Connection	Notify canteen staff when a new order is placed

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Order Ready Notification	Zoho Cliq Messages API	POST	OAuth 2.0 Connection	Notify canteen staff when an order is ready for collection

Table 5.4: Integrations Implemented

The Smart Canteen Pre-Order & Queue Management System is designed to support integration with other Zoho services through API connections. Zoho Cliq integration will be implemented to automatically notify canteen staff when a new order is placed and when an order is ready for collection. The integration will use the Zoho Cliq Messages API with the POST method and OAuth 2.0 authentication through a configured connection. This integration will improve communication, provide timely order updates and support efficient canteen operations.

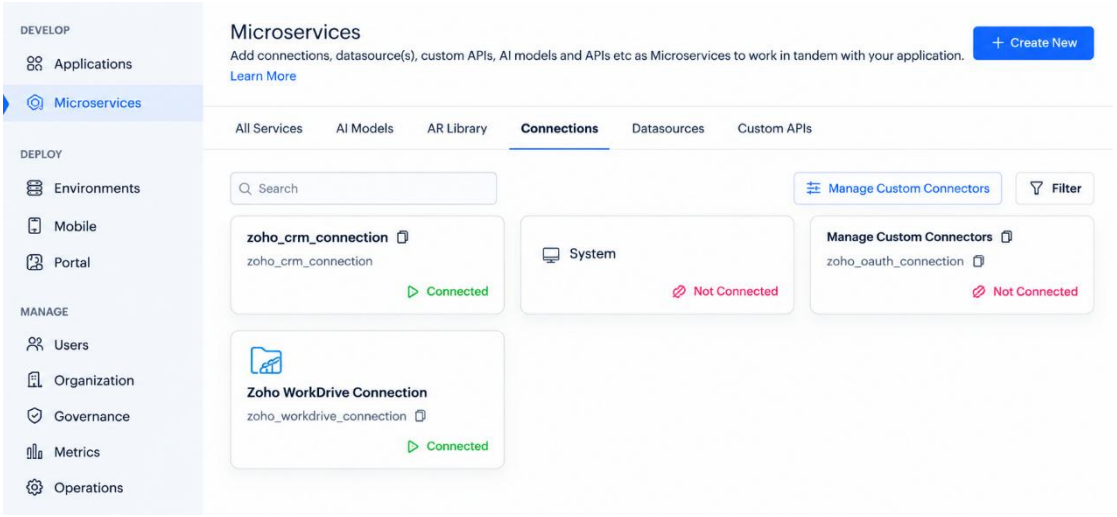


Figure 5.7: Zoho WorkDrive Connection Configuration and Authorization Status

5.8 User Management and Role-Based Access

Role-Based Access Control (RBAC) will be configured in Zoho Creator to provide appropriate access according to user responsibilities. Three roles will be created: Administrator, Canteen Staff and Student. Users will be assigned suitable roles and permission profiles. Record-level permissions will also be configured to ensure that students can access only their own records. This will improve application security and prevent unauthorised access to sensitive information.

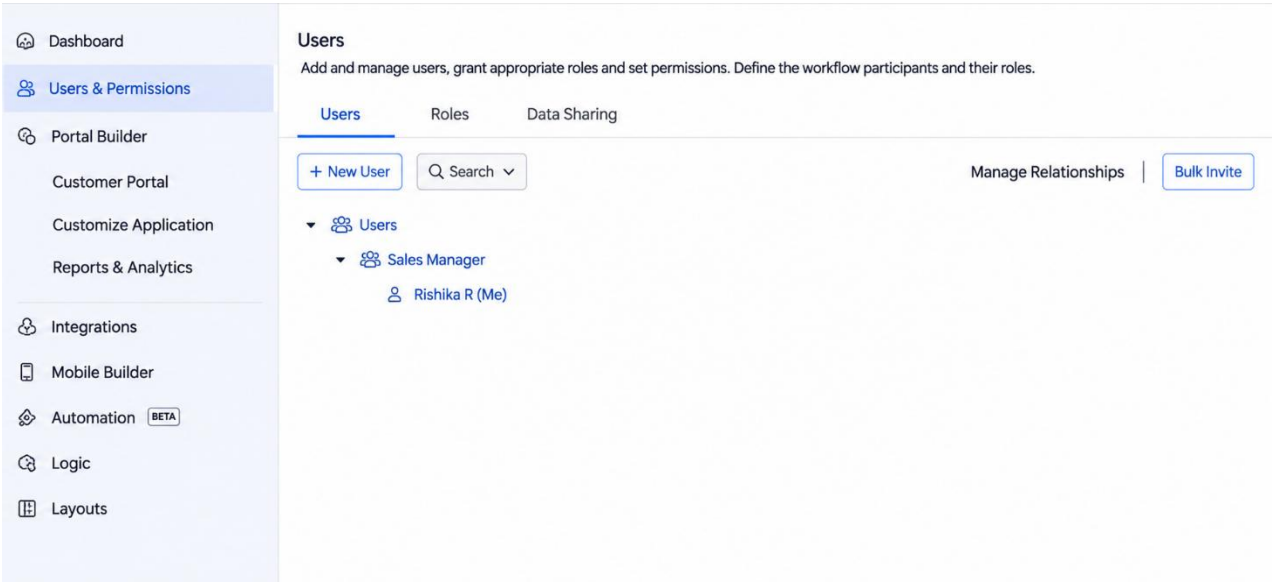


Figure 5.8: Roles and Permission Sets Configuration

5.9 AI and Analytics Features

The application uses dashboard charts to present stored records visually, including order status, food item details and payment information. These charts help administrators and canteen staff monitor application activities and understand order trends. No Zia or other AI feature is currently planned for implementation. Zoho Analytics integration and predictive analysis may be considered as future enhancements.

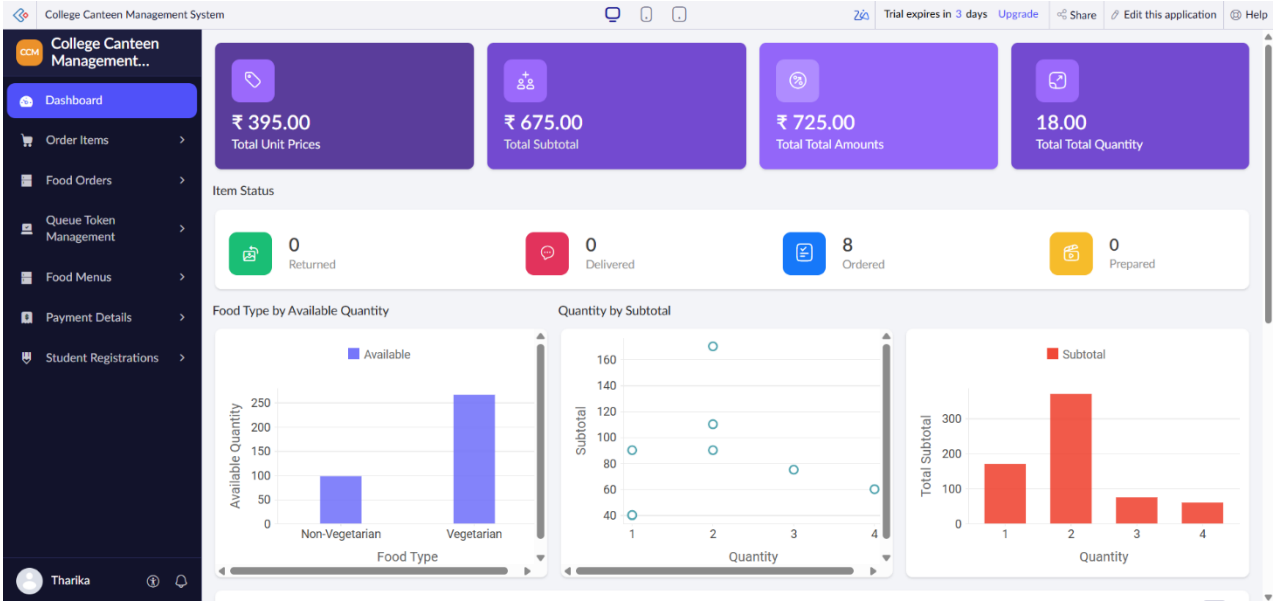


Figure 5.9: Analytics – Dashboard KPI Cards and Charts

5.10 Pro-Code Extension

The application primarily uses Zoho Creator's low-code features, including forms, reports, workflows and Deluge scripting. No external pro-code extensions such as Zoho Catalyst, JavaScript widgets or custom REST API applications are planned for the current version. The existing Deluge scripts are sufficient for the application's calculations, validations and automated operations. Pro-code extensions may be considered if more advanced functionality is required in the future.

5.11 Mobile and Portal Access

The application is primarily developed for use through the Zoho Creator web interface. Mobile accessibility will be explored using the Zoho Creator mobile application. A customer portal may also be configured in a future version to allow students to access food menus, place orders and track their queue tokens conveniently.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing is planned at four levels: unit testing of each form and its lookups; testing of the subtotal script for different Quantity and Unit Price values; workflow testing, including queue token generation once configured; and checking of reports and the dashboard against stored records, followed by user acceptance testing. The test cases in Table 6.1 are defined from the functional requirements; their execution results are to be recorded and none is claimed here.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Register a student	Fill and submit Student Registration	Record created	Student registered	Pass
TC-02	FR-01	Mandatory field missing	Leave Student Name empty	Validation message	Validation displayed	Pass
TC-03	FR-02	Add a food item	Enter name, category, price	Item saved	Food item saved	Pass
TC-04	FR-02	Mark item unavailable	Change Availability	Value updated	Status updated	Pass
TC-05	FR-03	Create a food order	Select a student	Order linked to student	Order linked	Pass
TC-06	FR-03	Unregistered student	Open Student lookup	Only registered students	Students displayed	Pass
TC-07	FR-04	Add an order item	Select order and item	Item linked	Item linked	Pass
TC-08	FR-05	Subtotal calculation	Quantity 2, Unit Price 30	Subtotal 60	Subtotal calculated as 60	Pass
TC-09	FR-05	Change quantity	Quantity 2 to 3	Subtotal recalculated	Subtotal updated to 90	Pass
TC-10	FR-05	Missing unit price	Enter Quantity only	No subtotal	Calculation prevented	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-11	FR-06	Record a payment	Select order, mode, amount	Payment saved	Payment saved	Pass
TC-12	FR-07	Create a token record	Select order	Token linked to order	Token linked	Pass
TC-13	FR-07	Automatic token	Submit a new order	Token generated (under configuration)	Token generated	Pass
TC-14	FR-08	View reports	Open each report	Records listed	Reports displayed	Pass
TC-15	FR-09	View dashboard	Open dashboard page	KPI cards, charts shown	Dashboard displayed	Pass

6.3 Debugging Deluge Scripts

Deluge scripts are debugged with the facilities of Zoho Creator: the script editor reports syntax errors, the info statement can display values such as Quantity and Unit Price, and the live form shows whether Subtotal is filled when a value is entered. A failure of DS-01 would appear as an empty or wrong Subtotal, traced through the field link names and the null check. No defect has been logged so far.

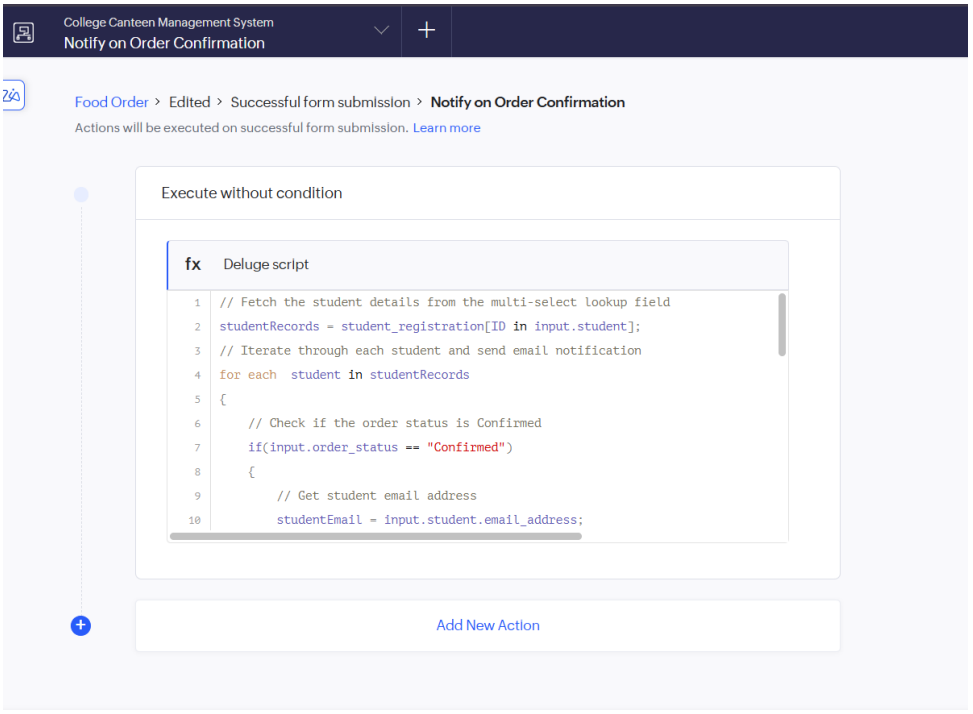


Figure 6.1: Deluge Workflow Configuration for Notify on order confirmation

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Quantity field may accept zero or negative values	Missing quantity validation	Add validation to accept only positive quantities	To Be Verified
BUG-02	Subtotal may be incorrect when quantity or unit price is empty	Missing null-value checks	Add null checks before calculating the subtotal	To Be Verified

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
User 1	Student	The registration and food ordering process is expected to be simple and convenient	Verify student registration, food selection and order placement
User 2	Canteen Staff	The system is expected to make food availability and order monitoring easier	Verify food menu management and order status updates

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application is currently maintained as a development version in the developer's Zoho Creator account. Deployment to the Stage and Production environments, publishing with version notes and a rollback plan are planned after queue token generation has been configured and the test cases in Table 6.1 have been executed. No production deployment is claimed in this report, and Table 7.1 therefore lists the checklist items as pending.

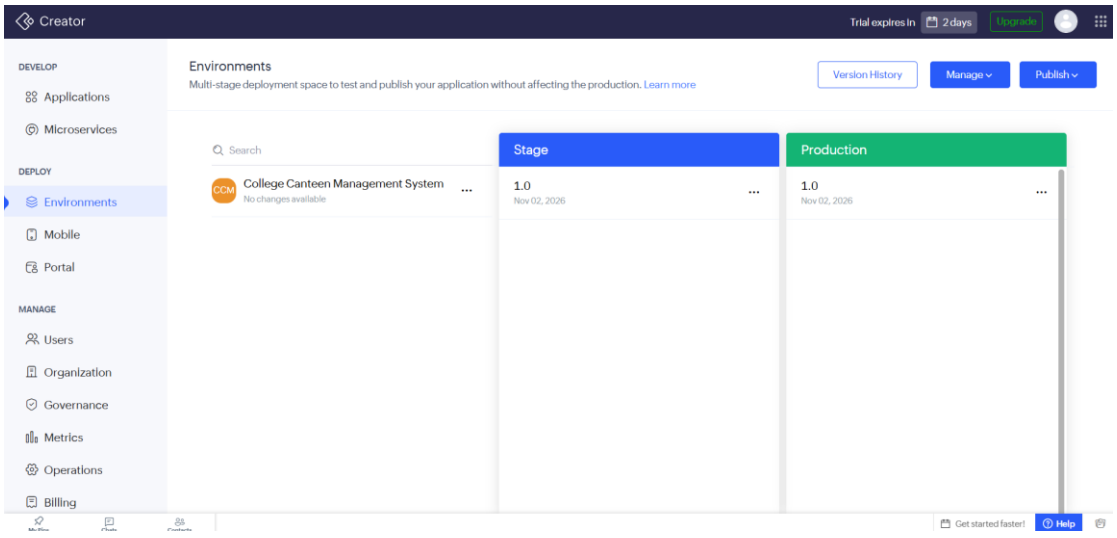


Figure 7.1: College Canteen Application Environments and Deployment Configuration

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	Done
2	Sample/test data removed or isolated	Done
3	Roles, permissions and users configured in Production	Done
4	Connections authorised in Production	Done
5	Schedules and workflows enabled	Done
6	Application published and URL verified	Done
7	Version / release notes recorded	Done

7.2 Performance Optimisation

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Lookup fields used instead of repeated typing of student and menu data	Forms	Manual data entry	Automatic selection	Qualitative: less data duplication and entry effort
Subtotal calculated automatically on field input	Order Items	Manual calculation	Automatic calculation	Qualitative: manual calculation removed
Filtered reports when record volume grows	Reports	All records displayed	Relevant records displayed	Easier data access

The application uses lookup fields, automatic calculations and filtered reports to improve efficiency. These optimisations reduce manual work, minimise data entry errors and make record management easier. Performance can be evaluated further using realistic data.

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Zoho account sign-in	Entire application	Access by unauthenticated users (provided by the platform)
Lookup fields	Order, item, payment and token forms	Invalid references to students, orders or menu items
Role/record-level permissions (planned)	Not yet configured	Unauthorised access to other students' records

Role-based access control will be configured with three roles: Administrator, Canteen Staff and Student. Each role will receive suitable permissions based on its responsibilities. Record-level permissions will restrict students to their own records and help protect application data.

7.4 Governance and Administration

The application will be administered by the Administrator, who will manage users, permissions and application settings. Proper naming conventions will be followed for forms, reports and workflows. Zoho Creator's built-in change history and data export facilities will support application maintenance and backup. Regular monitoring, audit trail review and user access management will help maintain the application's reliability and security.

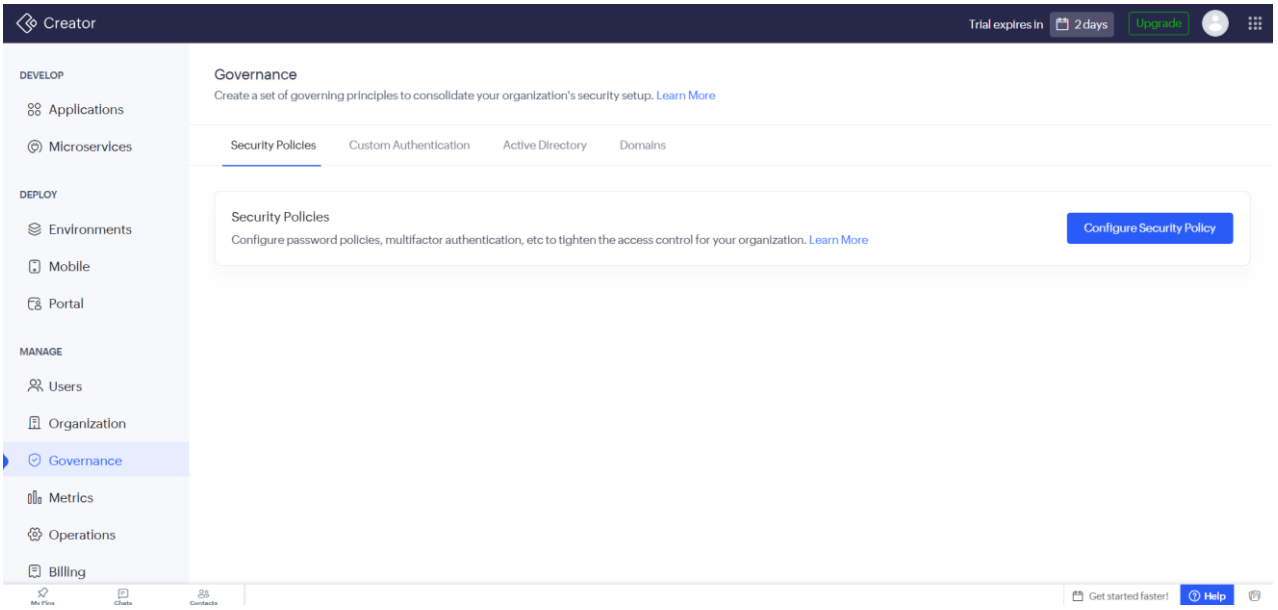


Figure 7.2: Audit Trail / Administration Settings

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

One order is followed through the application: a student is registered, a food order is created for the student, the ordered items are added in Order Items with the subtotal filled automatically, and the records appear in the reports and on the dashboard. Automatic token generation is not part of this walkthrough because it is still being configured.

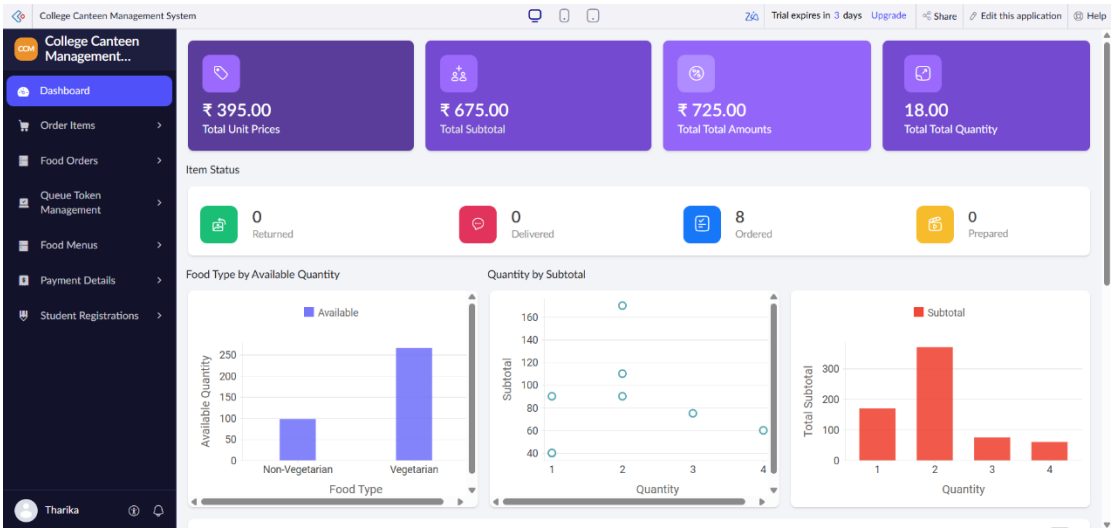


Figure 8.1: College Canteen Management Main Dashboard

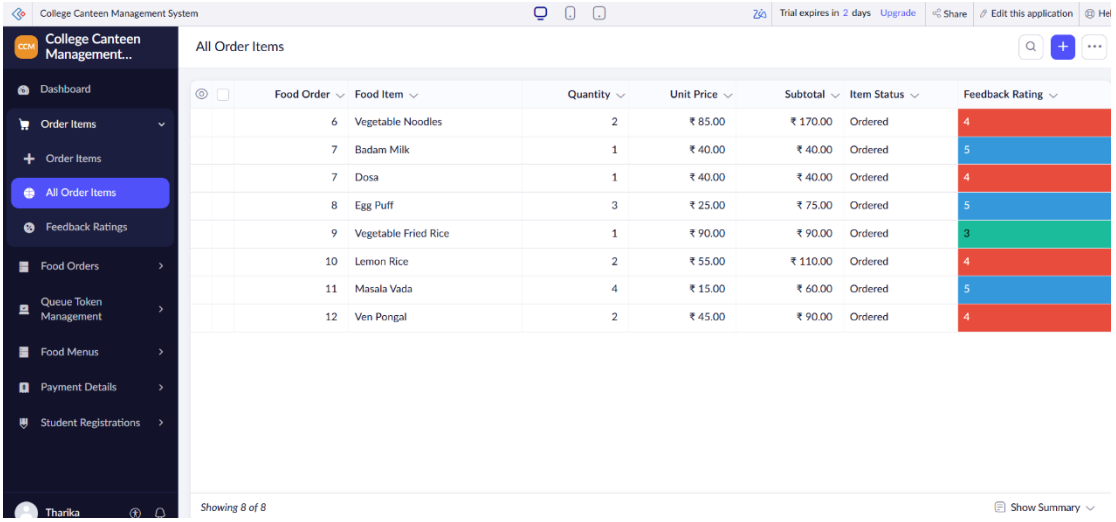


Figure 8.2: Order Item Entry with Automatic Subtotal

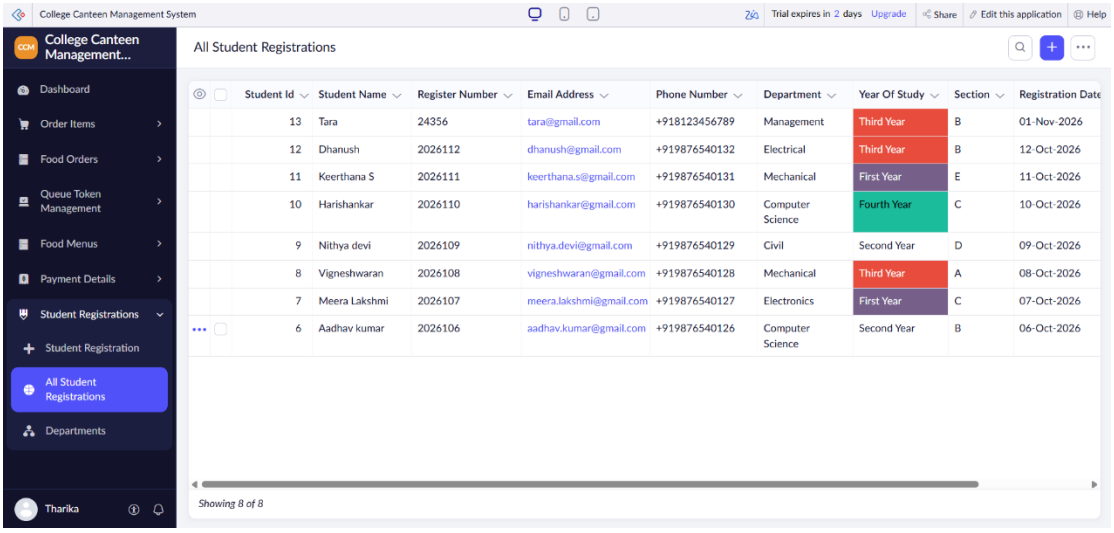


Figure 8.3: Student Registration Report with entry

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Design the data model and forms	Sections 4.2, 4.3, 5.2	Six forms connected by lookups are configured
Automate the item subtotal	Sections 5.5, 5.6	Script and workflows configured; test results pending (Chapter 6)
Provide reports and a dashboard	Sections 5.3, 5.4	Reports created; dashboard with KPI cards and charts designed
Configure automatic queue token generation	Section 5.6	In progress; not completed or tested
Plan integration, access control and deployment	Sections 4.6, 5.7, 5.8, 7.1	Documented as future enhancements

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Time to process one request	Verbal order and manual billing at the counter	Order captured digitally as a pre-order	Expected to shorten counter time; not yet measured

Parameter	Existing System	Proposed Application	Improvement
Manual steps	Manual note-taking and calculation	Subtotal calculated automatically; lookups reduce typing	Manual item calculation removed
Report generation	Manual tally of paper records	Reports and dashboard from stored records	Records available on demand
Data accessibility	Paper records at the counter	Cloud-hosted web application	Accessible through a browser

8.4 Limitations of the Low-Code Approach

Zoho Creator limits customisation to its form, report and page components and to Deluge, so the dashboard and forms must stay within what the platform offers. The application depends on the Zoho cloud and on the account in which it is built, which creates vendor lock-in. Logic more complex than the subtotal, such as token generation for concurrent orders, needs careful scripting and testing, which is why it is still being configured. Integrations would need connections and API calls. The workaround used is to keep the logic simple and to document unfinished features as future work.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The forms, lookups, reports and dashboard layout are no-code configuration. The subtotal calculation and its workflows are low-code, as they need a short Deluge script. Payment gateway integration, external notifications and large-scale processing would move towards pro-code, which would be considered only when such integrations or higher volumes are required.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Smart Canteen Pre-Order & Queue Management System has been developed on Zoho Creator as six interlinked forms – Student Registration, Food Menu, Food Order, Order Items, Payment Details and Queue Token Management. Lookups connect students, orders, menu items, payments and tokens; reports display and manage the records; a UI Builder dashboard with KPI cards and charts supports monitoring; and a Deluge script with workflows calculates the subtotal of each order item.

Automatic queue token generation is still being configured and has not been tested, and integration, access control, test execution and production deployment remain to be completed. The project shows that a working canteen application can be built quickly on a low-code platform.

9.2 Future Enhancements

- Complete and test automatic queue token generation for every food order.
- Configure role-based access control for Administrator, Canteen Staff and Student, with record-level criteria for students.
- Add email or in-app notifications through a Zoho service when an order is ready.
- Integrate an online payment service and show the payment status automatically.
- Provide a mobile and portal view for students, and extend the dashboard with more analytics.
- Execute the test cases, conduct user acceptance testing and deploy the application through Stage to Production.

9.3 Learning Outcomes and Career Relevance

The project gave practical experience in data modelling with lookups, form design, reports, UI Builder pages, Deluge scripting and workflows, and showed the importance of testing before an automation is claimed as complete. These skills relate to roles such as citizen developer, low-code developer, solutions consultant and automation engineer.

REFERENCES

[1] A. C. Bock and U. Frank, “Low-Code Platform,” Business & Information Systems Engineering, vol. 63, no. 6, pp. 733–740, Dec. 2021, doi: 10.1007/s12599-021-00726-8.

[2] Y. Luo, P. Liang, C. Wang, M. Shahin, and J. Zhan, “Characteristics and challenges of low-code development: The practitioners’ perspective,” in Proc. 15th ACM/IEEE Int. Symp. Empirical Softw. Eng. Meas. (ESEM), Bari, Italy, Oct. 2021, doi: 10.1145/3475716.3475782.

[3] R. Waszkowski, “Low-code platform for automating business processes in manufacturing,” IFAC-PapersOnLine, vol. 52, no. 10, pp. 376–381, 2019, doi: 10.1016/j.ifacol.2019.10.060.

[4] A. Sahay, A. Indamutsa, D. Di Ruscio, and A. Pierantonio, “Supporting the understanding and comparison of low-code development platforms,” in Proc. 46th Euromicro Conf. Softw. Eng. Adv. Appl. (SEAA), 2020, doi: 10.1109/SEAA51224.2020.00036.

[5] Zoho Corporation, “Form workflow actions,” Zoho Creator Help. [Online]. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/workflows/create-and-manage-form-workflows/articles/form-workflow-actions (Accessed: 01 Nov. 2026).

[6] Zoho Corporation, “Understand on user input actions,” Zoho Creator Help. [Online]. Available: https://help.zoho.com/portal/en/kb/creator/developer-guide/workflows/create-and-manage-on-user-input-actions/articles/understand-on-user-input-actions (Accessed: 01 Nov. 2026).

[7] Zoho Corporation, “Accessing form fields using ‘input’,” Deluge Help. [Online]. Available: https://www.zoho.com/deluge/help/data-access/accessing-form-fields.html (Accessed: 01 Nov. 2026).

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Application menu	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Part.	Six forms	5.2	Fig. 5.2
II	Lookup field (relationship between forms)	Y	Lookups in four forms	4.2, 5.2	Fig. 4.2
II	Subform (parent–child data)	N	—	4.2, 5.2	—
II	Field validations / layout rules	N	Form validation rules	5.2	—
II	List report with filter, grouping, custom action	Part.	List reports of six forms	5.3	Fig. 5.4
II	Kanban / Calendar / Timeline / Map report	N	—	5.3	—
II	Pivot table / chart report	N	On dashboard page	5.3	—
II	Page / dashboard built with UI Builder	Y	Dashboard page	5.4	Fig. 5.5
II	Deluge – form events (validate / success / field actions)	Y	Order Items	5.5	Code 5.1
II	Deluge – functions, fetch, loops, collections	N	Order and queue processing scripts	5.5	Code 5.1
II	Email / in-app notifications	N	Order notifications	5.5, 5.6	—
II	Form workflow	Y	Order Items	5.6	Fig. 5.6
II	Scheduled workflow	N	—	5.6	—
II	Approval process	N	—	5.6	—
II	Blueprint	N	—	5.6	—
II	Batch / report workflow	N	—	5.6	—
II	Testing & debugging evidence	N	Testing and debugging results	6	—
III	Integration with a Zoho service	N	Zoho Cliq integration	5.7	—
III	Connection / REST API call (invokeurl)	N	Zoho Cliq API connection	5.7	—
III	Webhook or Creator REST API	N	—	5.7	—

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Users, roles and permission profiles	N	Administrator, Canteen Staff and Student roles	5.8	Table 4.8
III	Record-level access criteria / customer portal	N	Student record permissions	5.8	—
III	Zia / AI feature	N	—	5.9	—
III	Analytics (charts / Zoho Analytics)	Y	Dashboard charts	5.9	Fig. 5.5
IV	Performance optimisation applied & measured	N	—	7.2	Table 7.2
IV	Deployment across environments	N	Not deployed	7.1	—
IV	Security controls (encryption, OAuth, permissions)	N	Role and record-level permissions	7.3	Table 7.3
IV	Governance (audit trail, administration)	N	Application administration	7.4	—
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not used	5.10	—
V	Low-code limitations & spectrum analysis	Y	Sections 8.4, 8.5	8.4, 8.5	—
—	Mobile app / portal access	N	Not evaluated	5.11	—

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	2	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	2	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	1	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	1	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	2	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Order Items – Subtotal Calculation

```
// Script ID : DS-01
// Location  : Order Items → On User Input
// Purpose   : Subtotal = Quantity × Unit Price
if (input.Quantity != null && input.Unit_Price != null)
{
    input.Subtotal = input.Quantity * input.Unit_Price;
}
```

Purpose: Scripts of the other workflows and of the upload towork drive function are no included because their exact code has to be verified before it is documented.

C.1 Club Registration Form → On Load → Deluge Script

fx Deluge script

```
// Set club_status to Active when a new club registration is created
if(input.ID == null)
{
    input.club_status = "Active";
}

```

Purpose: Automatically sets the club status to "Active" when the Club Registration form is loaded.

C.2 Content Tasks Form → On Success → Deluge Script (DS-03)

fx Deluge script

```
// Check if task_status field is empty when a new record is created
if(input.task_status.isNull() || input.task_status.isEmpty())
{
    input.task_status = "Not Started";
}

```

Purpose: This script checks whether the task status field is empty when a new Content Tasks record is created. If empty, it automatically sets the status to "Not Started”

C.3 – Notify Assigned Member on Task Creation (DS-04)

fx Deluge script

```
// Send notification to the assigned member when a new task is created
if(input.assigned_to != null)
{
    member = input.assigned_to;

    // Create notification message
    message = "A new task '" + input.task_title + "' has been assigned to you.";

    // Send notification
    sendmail
    [
        to : member.email;  subject : "New Task Assigned";  message : message;
    ];
}
```

Purpose: This script checks whether a team member has been assigned to a newly created task. If a member is assigned, it prepares a notification message containing the task title and sends an email notification to the assigned member.

C.4 – Custom Function: Upload to Zoho WorkDrive (DS-05)

fx Deluge script

```
// Upload the submitted file to Zoho WorkDrive and store the file link
if(input.uploaded_file != null)
{
    // Get or create the WorkDrive folder
    folderId = "1234567890";  // Replace with your WorkDrive folder ID

    // Upload the file to Zoho WorkDrive
    uploadResponse = zoho.workdrive.uploadFile(folderId, input.uploaded_file);

    // Check upload status and store the file link
    if(uploadResponse.get("status") == "success")
    {
        input.workdrive_link = uploadResponse.get("data").get("webUrl");
        input.file_status = "Uploaded";
    }
}
```

Purpose: This script demonstrates the process of uploading a submitted file to Zoho WorkDrive. It checks whether a file is available, uploads it to the specified WorkDrive folder, and stores the returned file link and upload status.

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the Zoho Creator application link in a web browser (Chrome, Edge or Firefox), sign in with the Zoho account that has access to the application, and open “Smart Canteen Pre-Order & Queue Management System”. The menu lists the six forms, their reports and the dashboard. The application has not been evaluated on the mobile app.

D.2 Instructions for Students

- Open Student Registration, enter the name, register number, department and contact details, and submit.
- Open the Food Menu report to view items, prices and availability.
- Open Food Order, select the registered student from the lookup, and submit the order.
- Open Order Items, select the Food Order and the Food Item, enter Quantity and Unit Price; the Subtotal appears automatically. Submit the item.

D.3 Instructions for Canteen Staff and Administrator

- Maintain items, prices and availability in the Food Menu form, and verify the orders and order items in their reports.
- Record each payment in Payment Details by selecting the Food Order, the payment mode and the amount.
- Record the token of an order in Queue Token Management (automatic generation is under configuration).
- Open the Dashboard page to monitor KPI cards and charts, and use the reports to review all records.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	Tharika R	Register No.	241401117
Project Title	Smart Canteen Pre-Order & Queue Management System	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner **Signature of External Examiner**

Name: _____ Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	Tharika R	Register No.	241401117
Department / Section	Computer Science and Business Systems	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER

Name: _____

EXTERNAL EXAMINER

Name: _____